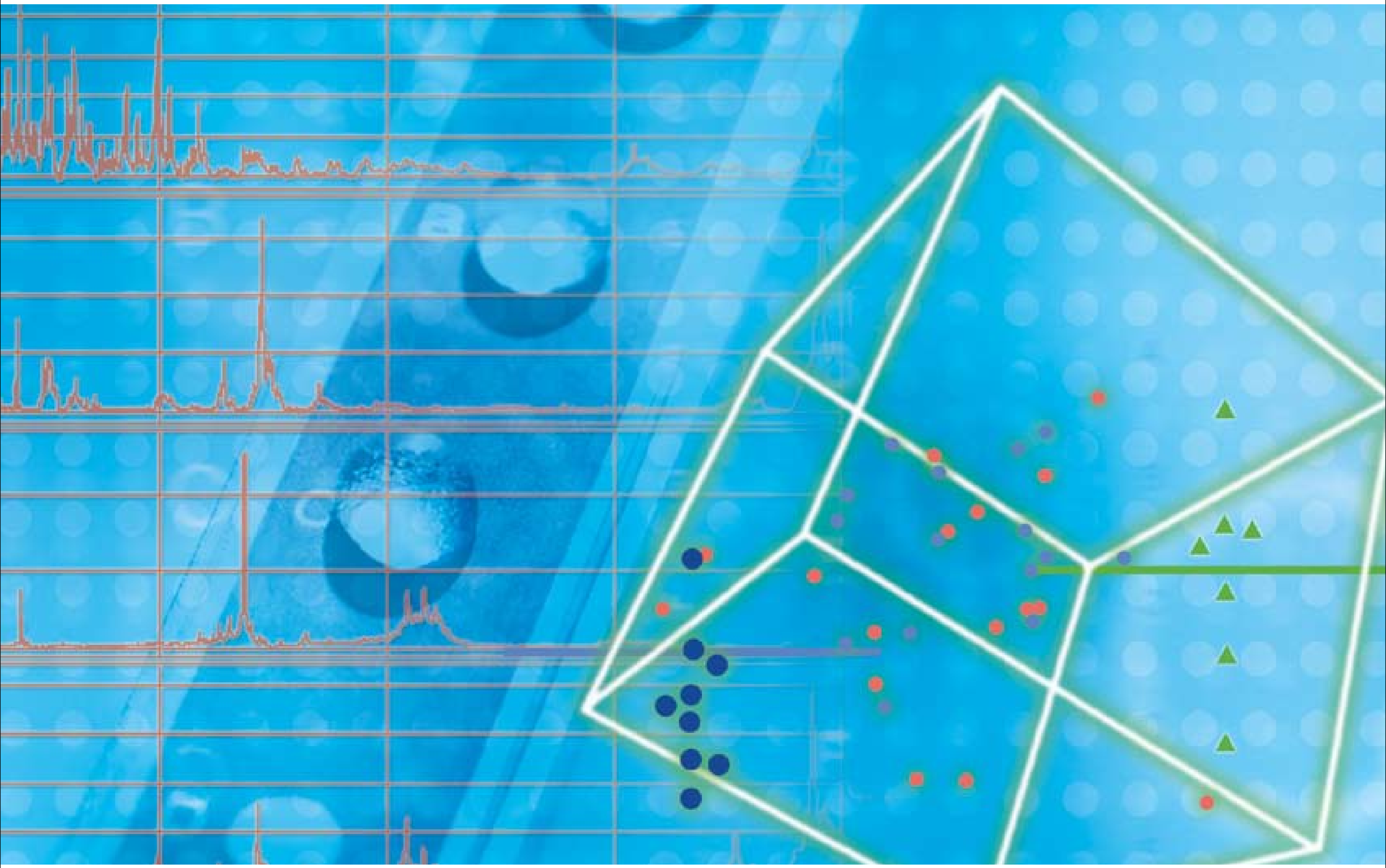

Biomarker Discovery Using SELDI Technology

A Guide to Successful Study and Experimental Design

BIO-RAD

Biomarker Discovery Using SELDI Technology

About This Guide

The goal of clinical proteomics research is to discover protein markers (biomarkers) and use them to improve the diagnostic, prognostic, or therapeutic outcome for patients or to assist in the development of novel drug candidates. For many of these applications, the use of various proteomics technologies and the recent emphasis on protein biomarkers have yielded a large number of candidate biomarkers; however, the small size and poor design of many studies have made validating these biomarkers challenging.

Biomarker attrition can be reduced with appropriate study designs that screen larger numbers of patients and take into account both preanalytical and analytical biases. The goal of this guide is to provide a series of recommendations for effective study design. General guidelines are provided that apply to the use of virtually any proteomics technology, and specific recommendations are also given for use of the ProteinChip® surface-enhanced laser desorption/ionization (SELDI) system.

A content map and summary of the main recommendations are provided at right.

- Part I offers an overview of the biomarker research process and the factors to consider when designing any biomarker research project, regardless of the technology used
- Part II discusses the process and recommendations to follow during study design, with an emphasis on factors that most influence data quality and reproducibility
- Part III offers specific recommendations and considerations for using the ProteinChip SELDI system; methods and protocols that have been optimized by the Bio-Rad Biomarker Research Centers for use with specific sample types are provided in Appendix A

Part I
The Biomarker Research Process

Study Design
Define the clinical question, samples, and workflow

Discovery
Detect multiple biomarker candidates

Validation
Select biomarkers with the highest predictive value

Identification
Purify and identify biomarkers

Clinical Assay Implementation
Design and implement biomarker-based clinical assays

Part II
Study Design

Clinical Question
- Ask a clear question that addresses a clinical need
- Determine the study type
- Define the success criteria

Sample Selection
- Select the model system and sample type
- Determine the appropriate sampling size
- Select appropriate controls
- Stratify the sample populations
- Determine inclusion and exclusion criteria
- Compile detailed sample annotations

Sample Collection, Handling, and Storage
- Implement standard methods for sample collection and handling
- Avoid systematic bias associated with collection site
- Freeze samples uniformly and avoid repeated freeze-thaw cycles

Experimental Design
- Understand the unique requirements of each study phase
- Define the general workflow for each phase
- Define the timing of the phases

Part III
SELDI Experimental Design

Assay Design
- Define the workflow
- Select the samples, controls, and standards
- Select appropriate ProteinChip SELDI array chemistries
- Select the methods of data analysis

Sample Preparation
- Use consistent and appropriate liquid-handling techniques
- Define protocols for the initial processing of samples
- Use fractionation and depletion techniques to improve detection of low-abundance proteins

Array Processing
- Determine the sample layout for each array
- Optimize sample dilution and buffer composition
- Standardize the methods of matrix application

Data Collection
- Perform regular instrument maintenance and calibration
- Optimize the acquisition protocols
- Acquire data using default processing parameters

Data Analysis
- Ensure proper annotation of spectra
- Process spectral data
- Group spectra into folders
- Evaluate the quality of the data
- Detect, label, and cluster peaks within one condition
- Perform univariate statistical analyses
- Perform multivariate statistical analyses

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The SELDI process is covered by U.S. patents 5,719,060, 6,225,047, 6,579,719, and 6,818,411 and other issued patents and pending applications in the U.S. and other jurisdictions.

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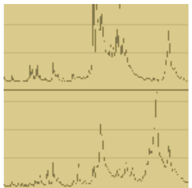
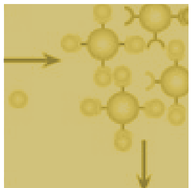
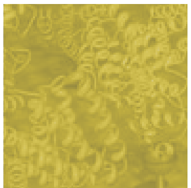





Part I: The Biomarker Research Process

The goal of biomarker or clinical proteomics research is to discover protein markers and use them to improve the diagnostic, prognostic, or therapeutic outcome for patients or to assist in the development of novel drug candidates. This section details the various phases of a biomarker research project and offers general recommendations that should be followed regardless of the proteomics technologies used.

Overview

Biomarkers are generally discovered through differential expression analysis, which is the determination of protein expression levels as influenced by disease state, therapy, or other differences between sample cohorts. Once differentially expressed proteins are identified, their expression levels can be used to classify organisms, individuals, disease states, metabolic conditions, or phenotypic responses to environmental or chemical challenges.

Five phases compose the path from study design to clinical application, and each phase presents different goals and requires unique experimental approaches (Table 1):

- Study design — in this initial phase, the objective is to detail the clinical question being asked and the types and number of samples, experimental workflow, and technologies that will be used. A successful biomarker research program begins with careful study design and implementation
- Discovery — in this phase, the objective is to find candidate biomarker proteins. For this purpose, a large number of conditions are screened to detect the maximum number of proteins and enrich low-abundance proteins. Samples must be chosen carefully and in sufficient quantities to provide statistical significance. Proteins

The Biomarker Research Process

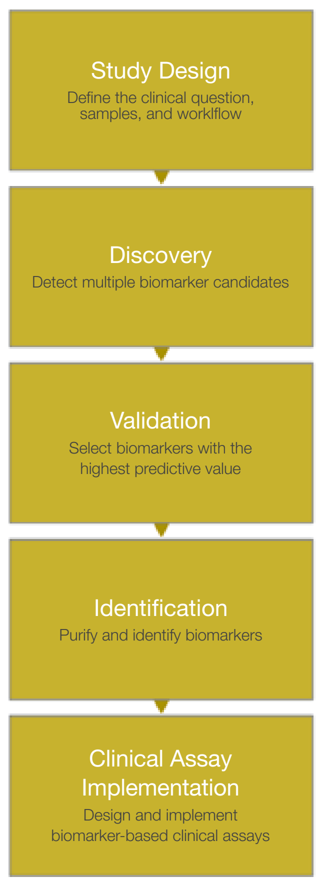


Table 1. Phases of a biomarker research project.

Phase	Goal	Number of Samples and Proteins Monitored
Study design	Define the clinical question, select and collect the appropriate samples and controls to address the clinical question, and select the proteomics platform(s) that will be used in each phase of the project	—
Discovery	Find candidate biomarker proteins by using a well-defined sample population and screening a large number of experimental conditions for the reproducible detection of the maximum number of proteins	10s of samples 1,000s of proteins
Validation	Assess the validity of candidate biomarkers against a larger, more heterogeneous population using a reduced set of experimental conditions	100s to 1,000s of samples 100s of proteins
Identification	Purify and positively identify candidate biomarkers	Varies
Clinical assay implementation	Develop and apply chromatographic or antibody-based assays that are optimized to provide robust, sensitive, and quantitative protein biomarker assays	1,000s of samples 10s of proteins

exhibiting statistically significant group- or time-dependent differences are described as candidate biomarkers and can be used alone (univariate analysis) or in combination (multivariate analysis) to generate predictive models

- Validation — the objective of this phase is to assess the validity of a marker against a larger, more heterogeneous population. The robustness of the candidate markers is tested against a level of biological variability that more accurately reflects the variability in the target population. This phase can either repeat and confirm the findings from the discovery phase on a larger sample set, or it may explore different variables that may affect the validity of the markers for a large population and, ultimately, the clinical utility of the biomarkers themselves
- Identification — in this phase, the most promising markers are enriched and purified; purified proteins are subsequently positively identified by peptide mapping or sequence analysis
- Clinical assay implementation — this phase can be performed at multiple points in a study. Assays can be either chromatographic or antibody-based, both of which can be optimized to provide a robust, sensitive, and quantitative assay

The order in which the five phases are completed depends on multiple factors, including the identity of the protein, availability of antibodies, and availability of appropriate samples for validation. The drug development time line can also influence the order of the phases. While the discovery phase is ideally followed by a comprehensive validation study, the design and completion of subsequent clinical trials often require extensive periods of time. It is, therefore, common to proceed directly to the identification phase following discovery. Identification facilitates the development of analyte-specific assays and provides insight into the biological mechanisms and pathophysiological process of the disease under study. The ProteinChip SELDI system, however, allows rapid and efficient validation of candidate biomarkers from large numbers of samples, increasing the statistical significance of any potential biomarker before identification.

Regardless of the final order of the validation and purification steps, a successful biomarker research program always begins with careful study design.

General Guidelines

Current proteomics and genomics technologies are extremely sensitive and can detect very small changes in expression levels. Some of these changes may stem from the biological differences related to a disease or treatment of interest. Others, however, may reflect the heterogeneity of patients across multiple sites, the inherent biological complexity and diversity of different sample types, and even small differences in the sample collection, processing, and analysis techniques used by multiple operators across multiple locations. As a consequence, results may be site-, study-, population-, or sample-specific and not of clinical use (Baggerly et al. 2004, Fung and Enderwick 2002, Hu et al. 2005, Mischak et al. 2007, White et al. 2005).

Reproducible results are achievable when sources of variability and bias are minimized (Semmes et al. 2005).

When bias cannot be avoided, the following can help minimize its effects on biomarker selection:

- Careful annotation of samples
- Randomization of samples
- Use of appropriate data analysis methods

The following guidelines affect the overall design and execution of any clinical research project. Though specific examples are given for the ProteinChip SELDI system, these guidelines apply to the use of any proteomics platform for biomarker discovery.

Minimize Bias

The key to maximizing reproducibility in biomarker research is identifying and minimizing all potential sources of preanalytical and analytical bias (Mischak et al. 2007, Poon 2007). Sources of preanalytical bias include any systematic differences in the patient populations or sample characteristics, as well as in the procedures used for sample collection, handling, and storage (Table 2). Sources of analytical bias arise from

Table 2. Factors that impact preanalytical bias.

Patient Characteristics	
	Age, sex, ethnicity
	Disease subtype and/or severity
	Type of control (healthy or disease)
	Location of sample collection (single or multiple sites)
	Smoking status, diet, other risk factors
	Drug treatments
	Study inclusion and exclusion criteria
Sample Characteristics	
	Type (blood, serum, plasma, urine, cerebrospinal fluid, cell lysate, etc.)
	Number of individuals
	Source (banked or prospectively collected)
Sample-Handling Procedures	
	Archived vs. new samples
	Collection protocols (number of sites, procedure, timing, initial processing, type of anticoagulant, etc.)
	Storage procedures (time, temperature, freeze-thaw cycles, aliquoting, etc.)

differences in how the samples are processed and analyzed (Table 3). These factors can have profound effects on the outcome of a discovery study and, more importantly, on the ability to apply discovered biomarker candidates to broader populations in subsequent validation studies.

Preanalytical Bias

Minimizing preanalytical bias ensures that differences observed between the experimental and control samples reflect innate biological differences and not differences in sample populations or sample collection protocols. In most projects, preanalytical biases are more difficult to control than analytical biases because of the limited number of samples often available for early studies. In addition, the sites, patients, and techniques used for sample collection and handling may differ.

To minimize preanalytical bias:

- Define the clinical or biological question and select appropriate samples, including those for all the experimental and control groups
- For retrospective studies using specimens from a sample bank, evaluate patient and sample histories and establish rigorous criteria for sample inclusion and exclusion
- For prospective studies, develop and apply standard operating protocols (SOPs) for all aspects of sample collection, handling, and storage
- Measure and document all potential sources of uncontrollable variation (Table 2) so that they are considered in the final data analysis

Table 3. Factors that impact analytical bias.

Sample-Processing Procedures	
	Liquid-handling methods (automated or manual, technique, equipment, etc.)
	Processing steps (denaturation, buffer components, delipidation, etc.)
	Fractionation and depletion methods
Experimental Protocols	
	Array types
	Sample load and placement on arrays
	Sample binding and washing procedures
	Matrix addition (type and method)
	Instrument settings
	Number of instruments, locations
Data Analysis Methods	
	Spectrum processing (baseline subtraction, noise reduction, normalization, etc.)
	Peak labeling
	Feature selection, statistical analyses
	Classification approaches

Analytical Bias

Minimizing analytical bias maximizes the discovery of true biological differences from a properly selected sample set by minimizing differences in how the samples are processed and analyzed. Analytical bias can be controlled largely through rigorous training, instrument qualification, and the use of SOPs.

To minimize analytical bias:

- Use sufficient numbers of replicates. At the minimum, use duplicates of each sample
- Randomize the order and placement of samples (for example, into a 96-well microplate format) during sample and array preparation. This ensures that any inadvertent or unavoidable sources of bias (such as variability in liquid handling, instrument fluctuations, or differences in array or reagent quality) affect both kinds of samples equally and are not represented as biological differences
- Process all samples together — including controls — and use reagents from the same lot if possible. Use automated liquid-handling and processing systems to help minimize variability
- Optimize the protein load and binding and wash buffers used for array preparation
- Maintain optimum instrument performance through operational qualification and regular detector calibration
- Account for lab-to-lab, instrument-to-instrument, and assay-to-assay variability as well as instrument drift over time by including reference and quality control samples for intensity normalization and comparison of relative peak quantitation
- Optimize data collection parameters for each experimental condition (fraction, array chemistry, matrix, and mass range)
- Directly compare peak intensity data collected for only one experimental condition using a single set of instrument parameters
- For multivariate analyses, combine biomarker candidates from different experimental conditions to improve diagnostic performance
- Analyze all data using consistent parameters for spectrum processing, feature selection, and statistical analyses
- Maintain detailed records of all sample-processing and data analysis steps for each sample so that any differences are taken into account during data analysis

Implement Standard Operating Procedures (SOPs)

Effective use of any proteomics platform in biomarker research requires training of all personnel and adherence to strict guidelines and protocols. SOPs are critical to implementing the highest operating standards for quality and reproducibility (Baggerly et al. 2004, Hu et al. 2005, Mischak et al. 2007). SOPs also facilitate the validation of biomarkers by other groups using different sample sets. Taking the time to establish and optimize SOPs for sample collection, sample handling, sample preparation, array processing, and data acquisition and analysis helps minimize both preanalytical and analytical bias and increases the opportunity to discover robust biomarkers.

Collaborate With Specialists

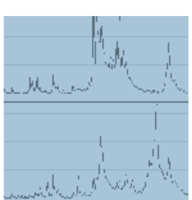
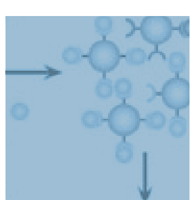
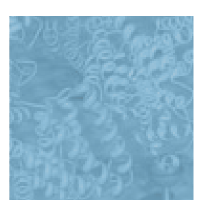
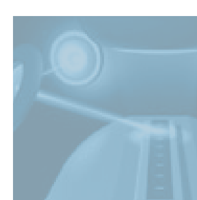
For proteomics-based clinical proteomics research, study design entails defining a clinical question, selecting appropriate patients and samples for analysis, and carefully planning a series of experiments that generate reproducible results. Since these considerations require expertise in a number of fields, it is often advantageous to design and conduct studies in conjunction with a group of specialists. These specialists might include, for example, a clinician, proteomics researcher or mass spectrometry technician, and biostatistician. In addition, it may be helpful to include bioinformaticists, epidemiologists, clinical and analytical chemists, and biologists or biochemists during appropriate phases of the project. Such a collaborative effort is most critical during study design and data analysis (Clarke et al. 2005, Mischak et al. 2007).

Develop Data Analysis Strategies Before Acquiring Data

Mass spectrometry-based proteomic profiling techniques generate many peak intensity features per sample, significantly more than the total number of samples in a study. This results in high-dimensional data, which carry a higher risk of false discovery and over fitting of multivariate models. Therefore, the involvement of a biostatistician is recommended for both the study design and data analysis phases of a project.

Employ assistance from a biostatistician to calculate the number of samples required for statistical relevance and to plan data analysis strategies that minimize the effects of the two most common mathematical hurdles: over fitting and random solutions. Over fitting occurs when a mathematical model is so tightly tuned to correctly classify the training data that it performs poorly on broader sets of data. Random solutions, or false discovery, of biomarkers occur when high-dimensional data create statistically significant solutions by pure chance. The expertise of a biostatistician may also be used to develop solid statistical assumptions and to apply conservative feature selection and statistical cross-validation within a sample set. If possible, the analysis should also be tested using an independent validation data set.

Part II: Study Design

Part II: Study Design

This section provides guidelines for planning the general aspects of a biomarker discovery project. It addresses the steps that should be undertaken before sample separation and analysis and provides recommendations for minimizing preanalytical bias.

Clinical Question

Ask a Clear Question That Addresses a Clinical Need

Most successful studies begin with a clear, narrowly defined clinical question. Broad questions, though possibly applicable to a wider population, generally require larger sample sets and introduce more variables, making them more complex to validate (Mischak et al. 2007). An example of a broad research question is one that aims to identify biomarkers indicating the early stages of a particular form of cancer. Such a question implies large sample numbers, as it requires screening the general population (and many years of experimentation) for validation. A more narrowly defined version of the same question, on the other hand, might aim to identify biomarkers predictive of cancer in patients with a suspicious mass or other definable sign; this would require screening smaller sample sets while maintaining diagnostic utility.

The clinical question should specify a measurable result of clinical utility and aim to yield results that improve current diagnostic, prognostic, or therapeutic methods. For example, a study might generate results that improve patient management and outcome by yielding a new test or therapy that is less invasive, less expensive, or more effective (for example, by providing earlier disease detection). This requires consideration of the disease or metabolic pathway under investigation, any existing therapies and markers, and the type of impact the expected results might have on the current methods of diagnosis, prognosis, or treatment. To ensure the biomarker discovery project is of clinical value, define the question and desired outcome with the involvement of a clinician who has experience with the disease or treatment under investigation.

Determine the Study Type

Two types of studies are generally used (Figure 1):

- Retrospective studies use patient or animal model samples from a sample bank for which the clinical outcome is already known. Retrospective studies often rely on information collected by questionnaire, case records, or sample banks — often years after the patients have developed the disease
- Prospective studies enroll patients or define model systems and monitor their progress, symptoms, or disease development over time

It is often easier to obtain banked samples with the associated patient information than to initiate a new study. As a result, biomarker discovery studies are almost always retrospective, with various levels of rigor in the selection of controls. One of the biggest problems with retrospective studies is that patient information may be inaccurate, incomplete, or hard to obtain — all of which introduces greater potential for preanalytical bias.

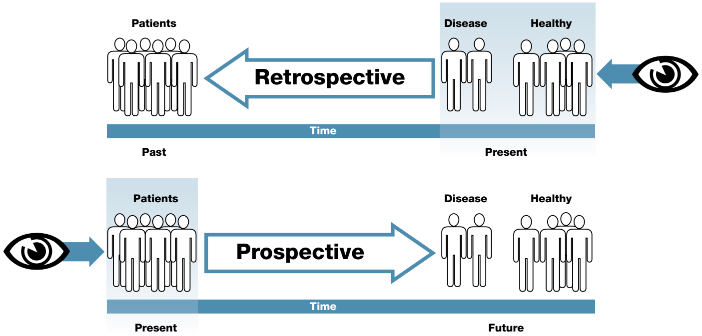


Fig. 1. Differences between retrospective and prospective studies. In retrospective studies, the outcome of patients is known at the time the study is initiated, while in prospective studies, patients are followed over time. Patient information and samples are often easier to obtain for retrospective studies; however, for use in screening or early detection tests, candidate biomarkers identified in retrospective studies must be validated in prospective studies.

In addition, retrospective studies are often based on a different clinical question generated some years prior to the current study; as a result, they often do not include an optimum set of controls.

For clinical applications involving early detection or screening, biomarkers obtained in retrospective studies must be validated in prospective studies.

Define the Success Criteria

Success criteria for a clinical proteomics question depend on how any available "gold standard" for clinical decision-making performs by four test characteristics (see sidebar, below). In instances where no such standard exists, use the expertise of a clinician to determine the statistical requirement for accepting and adopting the findings or any resulting clinical assay.

Test Characteristics

The utility of diagnostic tests is determined using four test characteristics: sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). The table at right shows how these characteristics are related to actual test results (letters a–d denote numbers of samples or patients in a category).

Test Result	Disease Present (+)	Disease Absent (–)	Totals
Positive (+)	a	b	a + b
Negative (–)	c	d	c + d
Totals	a + c	b + d	

Sensitivity — rate of true positive detection; rate at which the test detects disease or condition in a person known to have it; equivalent to a/(a + c)

Specificity — rate of true negative detection; rate at which the test does not detect disease in a person known to not have it; equivalent to d/(b+d)

PPV — chance that a person who tests positive actually has the disease or condition; equivalent to a/(a+b)

NPV — likelihood that a negative result is accurate; equivalent to d/(c+d).

For most screening tests, sensitivity and specificity are fixed, while PPV and NPV depend upon the prevalence of the disease in a population. For any test to be useful, it should have high sensitivity and specificity; for screening purposes, the test should also have a high PPV to avoid the possibility of high rates of false positives (which might lead to unnecessary, expensive, and potentially traumatic follow-up procedures). In any case, the desired test characteristics should be estimated as accurately as possible before any test is used in a clinical setting.

Single biomarkers rarely provide sufficient PPV. Indeed, combinations or panels of biomarkers often have much better performance characteristics. For this reason, most biomarker studies these days strive to screen larger numbers of samples and identify panels of biomarkers for use in clinical assays.

Sample Selection

Select the Model System and Sample Type

The choices of model system and sample type ultimately depend on the question being investigated. For example, is the ultimate goal of the research to better understand the mechanism and progression of a disease (and so involves tissue or cell isolation), or is it to generate a new, less invasive clinical assay involving biological fluids (such as whole blood, serum, or urine)?

Clinical proteomics research uses a number of model systems, including human patients, animal models (such as primate or murine models), and even cell culture or in vitro systems. Whereas human patients ultimately represent the most accurate model for clinical studies, nonhuman models display less biological variability and offer the opportunity for experimentation in different or recombinant genetic backgrounds. In some cases, such as toxicity studies, experimentation in humans may be impractical, unnecessary, or even unethical; in such cases, animal or in vitro models may serve as effective options.

For ProteinChip® SELDI analyses, a variety of sample types are amenable to profiling, including biological fluids such as serum, plasma, cerebrospinal fluid (CSF), and urine, as well as tissue and cell extracts, cell culture medium, and cells collected by laser capture microdissection. These sample types may be taken from human patients, animal models, or in vitro models. Serum and plasma are commonly used for proteomics studies due to the relative ease and minimal invasiveness of their acquisition and their general diagnostic use. Appendix A of this guide contains protocols developed for ProteinChip SELDI analysis of serum, plasma, CSF, and urine samples.

Determine the Appropriate Sampling Size

General considerations for determining the sample size for the discovery phase are given here; for more specific recommendations for the different phases of a SELDI study, refer to Table 4 in Part III of this guide.

Estimate the number of samples required to attain statistical significance. Use statistical power calculations that take into account the complexity of the clinical question, the biological variability of the organism being studied, the magnitude of change in marker concentration, analytical reproducibility, phase of the study, incidence of the disease, and other factors (Lenth 2001 and 2007, Mehta et al. 2006). The help of a biostatistician can be invaluable in performing these critical calculations.

Cell cultures and other in vitro model systems, which are highly homogeneous and controlled, require relatively small sample sets (as few as 2–6, but defined by the specific study) and small numbers of replicates to yield informative biomarkers in SELDI-based biomarker discovery studies. Animal models display differing amounts of interanimal variability, which affects the number of samples recommended to attain statistical significance. For example, transgenic models are generally very homogeneous and require a relatively low number of animals (typically 8–10 animals) in each treatment or disease group, while nonhuman primates show variability similar to that observed in human subjects and, therefore, should be subjected to similar guidelines as human patients.

Biomarker discovery in human samples is more challenging. Since the human proteome is extremely complex and the proteomic profiles are subject to environmental variability (time of day, meals, etc.) and preanalytical bias (site of collection, storage, etc.), researchers working with human samples need a large sample set to overcome patient-to-patient variability. Given the inherent variability in human populations, the only viable way to increase the statistical power of clinical studies is to increase the number of patient samples in the discovery phase.

For SELDI-based, discovery-phase human studies, we recommend a minimum sample size of ~30 per classification group (for example, treated vs. untreated, or disease vs. control) to accommodate the high biological variability of the human population. This number of samples generally offers >90% statistical confidence in single markers (with P values of <0.01) and the ability to use some forms of multivariate analysis. Use of samples from multiple collection sites or the study of complex clinical questions (for example, diseases that arise by multiple or unknown pathways, such as cancer, Alzheimer's disease, and depression) may require significantly larger sample sets. In validation studies, even larger numbers of samples are required (100–1,000 samples, depending on the complexity of the question and the panel of biomarkers), and they should be taken from a larger, more heterogeneous population.

Multiple Time Points

If collecting a large number of samples is challenging, consider including multiple time points or baseline or pretreatment time points. These permit analysis of time-dependent changes and are better suited to revealing consistent differences in populations that exhibit high variability. In such populations, the wide range and high degree of biomarker overlap between samples at any single time point may mask differences between the sample cohorts. Examining multiple time points enhances the detection of such differences and identifies possible trends.

In addition, absolute protein concentration can vary significantly between individuals; when changes in an individual are compared to a baseline, the reliance on absolute protein concentration is reduced.

Multiple Doses

If it is relevant to the study, also consider using multiple doses to characterize protein changes. A typical preclinical toxicity study might include at least three groups (samples from high- and low-dose treatment groups and vehicle controls) with three time points: pretreatment, an early time point, and a late time point. In most cases, toxicity is only detectable by conventional assays at the late time point or at the higher dose. The pretreatment time point serves as a baseline for measuring subsequent changes in protein expression, and the early time point is assessed for the capability to identify toxicity earlier than using conventional toxicity endpoints. Samples from patients treated with high doses can be used to select candidate biomarkers, which are then evaluated at the lower dose.

Select Appropriate Controls

As with any scientific investigation, the selection of experimental controls for a SELDI-based biomarker discovery project is as important as the selection of patients or experimental samples. Experimental controls can be internal (taken from the same individual, but at a baseline time point or from surrounding healthy tissue) or external (taken from other individuals in a population). Often, using more than one type of controls adds to the confidence in the resulting data (Mischak et al. 2007). It is seldom effective to simply compare proteomic data from a group of diseased individuals only to a group of healthy or "normal" individuals (healthy controls). If possible, use an additional set of controls composed of samples collected from patients with other diseases or disorders that have clinical profiles mimicking those of the disease under study; for example, patients with a disease affecting the same organ (disease controls). In many cases, the types of samples analyzed may be different in the discovery and validation phases of biomarker research.

If obtaining samples from a sample bank, ensure that the samples and controls were collected and handled as similarly as possible. In addition, collect all the clinical information available for every sample and use this information to determine the suitability of the samples for inclusion in or exclusion from the study.

For details about the other types of controls used in SELDI studies, refer to the Assay Design section of Part III of this guide.

Stratify the Sample Populations

Stratification is the process of grouping members of a population into relatively homogeneous subgroups before sampling. Since more homogeneous subgroups usually yield more statistically significant biomarkers, consider the sample or patient characteristics listed in Table 2 in selecting and stratifying samples for a study. During pilot studies and the discovery phase of a project, hold as many of these parameters constant within a subgroup as possible, since all analysis of the resulting data is based on the group to which a sample is assigned. For validation of candidate biomarkers and for any result to be clinically relevant, however, extend the studies to patients of a range of ethnicities, ages, and other appropriate characteristics so that they represent a true clinical population.

Determine Inclusion and Exclusion Criteria

Inclusion must take into account the suitability of the population for addressing the clinical question, including considerations such as the specific diagnostic and prognostic criteria. Exclusion criteria can be based on the lack of suitability to address the defined clinical question as well as safety concerns that may put the health of the patient at risk; however, consider the impact of exclusion on the generalization of the study during data analysis and study summation. In addition, consider the accuracy of diagnoses — the disease and its symptoms, quality of annotation (described in the next section), etc.

Compile Detailed Sample Annotations

Sample annotation refers to all of the descriptive information associated with a given sample. Just as in sample stratification, all data analyses rely on the information contained in sample annotations, so it is vital that all annotations are complete, correct, and unambiguous.

Begin to compile sample annotations prior to the start of experimental work. In addition to any unique identifiers, a clear and complete description of the samples and any prior sample treatment procedures is necessary. ProteinChip data manager software, Enterprise Edition, includes a "virtual notebook" function that makes incorporating sample annotation into the spectra quick and easy. See the ProteinChip Data Manager Software Operation Manual for details.

Sample Collection, Handling, and Storage

Robust results require the careful and consistent application of all treatments to every sample, in every sample class, from the time of sample collection to the point when proteins are introduced for analysis. For this reason it is vital to establish standardized protocols for sample collection, handling, and processing as part of the design of any study (refer to Table 2 for variables to consider). If obtaining samples from a sample bank, obtain samples and controls that were collected and handled as similarly as possible and consider all the clinical information available to decide upon the suitability for inclusion in or exclusion from the study.

ProteinChip SELDI technology is compatible with most salts and detergents, so the guidelines for sample collection and handling revolve primarily around consistent application of protocols rather than a single technique or buffer composition. For more detailed recommendations for sample handling and processing, refer to Part III of this guide.

Implement Standard Methods for Sample Collection and Handling

Design a precise sample collection protocol and ensure that it is rigorously followed, particularly if samples will be collected at multiple sites or by multiple personnel. Maintain consistent:

- Timing of collection relative to the disease process (for example, pretreatment or postoperative)
- Types of equipment and reagents (for example, collection tubes and additives). Do not exchange or substitute consumables during the course of the study. Use consumables made of materials such as polypropylene that do not bind protein
- Choice of anticoagulants for plasma samples. For SELDI, EDTA and citrate have been used extensively and are generally preferred over heparin, as the highly charged nature of heparin can significantly alter the chromatographic behavior of proteins and the resulting ProteinChip SELDI profiles
- Timing between sample collection and any processing and transportation methods between the site of collection and the processing laboratory; for example, on ice for a maximum of 4 hours
- Methods and timing of all processing steps. The time limit for sample collection is less important than the consistency of the sample-processing procedure. Even a slight variation in sample processing (for example, time before freezing, centrifuge speed, etc.) can generate inconsistent results

For cell culture models, use of serum-free media eliminates the high background from fetal calf serum proteins. If serum-containing medium is required for cell viability, wash the cells thoroughly with a serum-free medium or buffer before preparation and profiling of the cell lysate. For biomarker discovery from culture medium, use appropriate media controls (for example, fresh or conditioned media, untreated cells, etc.) and multiple time points or doses to control for contaminating serum proteins.

Avoid Systematic Bias Associated With Collection Site

Even when care is taken to collect samples using the same protocol at different sites, there are often subtle differences between collection sites that sensitive technologies such as SELDI can detect.

To avoid systematic bias associated with collection site:

- Use multiple collection sites — even during the discovery phase — to improve the likelihood that initial results are later validated (Clarke et al. 2005). Though it may be tempting to decrease bias by collecting samples for small discovery studies at the same site, this may pose problems for clinical reproducibility when moving to multisite validation. In addition, use of a single site is rarely feasible in clinical trials, in which multiple sites are often involved
- Select sites with the largest numbers of samples with appropriate controls to allow for independent statistical analysis for each site and subsequent selection of common markers
- Collect control and disease or treatment samples from each of multiple sites, if possible. When multiple sites are involved, represent as many of the sites as possible in all disease/treatment groups
- Collect samples in a blinded fashion and in a random order. Ideally, collection personnel should not know which individuals are serving as controls and which are in the disease or treatment group

Freeze Samples Uniformly and Avoid Repeated Freeze-Thaw Cycles

Freeze all samples immediately after processing, preferably at –70°C, but at least at –20°C. Avoid prolonged (>4 hours) storage at 4°C. While transporting frozen samples, care must be taken to avoid thawing. During storage, samples must remain frozen at –70°C.

Avoid repeated freeze-thaw cycles by aliquoting samples prior to initial freezing. Repeated freeze-thaw cycles compromise the integrity of samples by generating oxidative modification and degradation products, which change resulting protein profiles. If freeze-thaw is unavoidable, subject all samples to the same number of freeze-thaw cycles.

Experimental Design

The last component of study design involves planning the overall experimental workflow — planning all aspects of performing the biomarker study, from selecting the appropriate proteomics platform to assay design, data collection (instrument settings), and analysis. An effective experimental design:

- Ensures that the study answers the clinical question being asked
- Minimizes all sources of analytical bias
- Tests the predictive value of the resulting biomarkers

For details and specific considerations concerning each aspect of the SELDI workflow, refer to Part III.

Understand the Unique Requirements of Each Study Phase

Five phases compose the path from biomarker discovery to clinical application, and each phase presents different goals and requires unique experimental approaches (see Part I). Though the order in which these phases are undertaken may vary, understanding the particular goal of each phase of the project and the anticipated results helps to ensure the selection of appropriate samples and analysis methods.

Each phase may require several iterations and optimization. For example, the discovery phase requires different degrees of optimization depending on whether standardized or validated protocols exist for the sample type being used.

Define the General Workflow for Each Phase

For each phase, determine the following:

- Which samples and sample preparation procedures will be used? Define liquid-handling techniques, initial sample-processing steps, and fractionation and/or depletion strategies for the enrichment of low-abundance proteins
- Which proteomics platform will be used? Different methodologies may be used for different phases of the workflow. The method selected should be the most appropriate for the question being asked and for the unique goal of each phase, but may also depend on available expertise and resources
- Which data analysis strategies will be used? These should be defined in conjunction with both the clinical study design and experimental workflow to maximize chances of success. Appropriate design requires incorporation of statistical principles and assumptions that will be used during data analysis

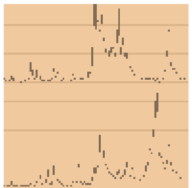
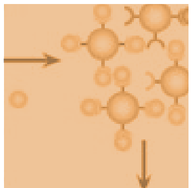
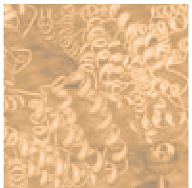
Define the Timing of the Phases

Define when and over what period of time each phase will take place. Anticipate the time course of each set of experiments and ensure that proper controls exist for instrument and sample variability over time.

For further advice on clinical study design, refer to the following web sites:

- FDA good clinical practice program (www.fda.gov/oc/gcp/default.htm and www.fda.gov/cder/guidance/959fnl.pdf)
- EU Directive 2001/20/EC (eur-lex.europa.eu/en/index.htm)
- Canadian Medical Association ICH guidelines (mdm.ca/cpgsnew/cpgs/index.asp)

Part III: SELDI Experimental Design


Part III: SELDI Experimental Design

The SELDI experimental workflow is divided into multiple steps: assay design, sample preparation, array processing, data collection, and data analysis. As with any proteomics technology, each step presents a number of variables that influence the quality of resulting data. To minimize analytical bias, take the time to optimize the operating parameters for each step of the workflow, establish detailed standard operating procedures (SOPs), and follow them rigorously.

This section provides an overview of the factors to consider when planning a biomarker research project using the ProteinChip SELDI system. It offers general recommendations and descriptions of the variables to consider for optimum results and reproducibility. Refer to Appendix A for sample and array preparation protocols that have been optimized by the Bio-Rad Biomarker Research Centers for use with plasma, serum, cerebrospinal fluid (CSF), and urine.

Assay Design

When planning a SELDI-based study, consider each phase of the project separately (Table 4). For each phase, determine the appropriate samples and controls to use, and plan the most effective workflow.

Define the Workflow

Define and optimize the different steps of the SELDI workflow for each phase of a project. For biomarker discovery and differential expression analysis, maximize the possibility of finding peaks of interest by determining the procedures for sample preparation, array processing, and data collection that maximize both the number of peaks detected and their reproducibility. Once peaks of interest are selected, optimize the same procedures prior to validation to maximize quantitative reliability, resolution, and signal-to-noise ratio.

Take the time to optimize the procedures used for the following:

- Sample preparation, including appropriate liquid-handling techniques, initial sample-processing steps, and fractionation or depletion strategies to enrich low-abundance proteins
- Selection and processing of ProteinChip arrays, with emphasis on optimizing sample dilution and buffer composition
- Data acquisition, with emphasis on optimizing laser energy
- Data analysis

More details about optimization strategies are presented in the sections that follow.

Table 4. Phases* of SELDI-based human clinical proteomics studies. The guidelines provided are based on experience working primarily with human plasma and serum samples. The number of samples required for statistical significance using other methods or sample types may vary and should be determined empirically (for example, smaller sample numbers are generally required for animal model and cell culture studies).

Phase	Number of Samples	Comments
Method development	10 to 40	Using a small number of samples or pooled samples, optimize experimental methods to maximize peak counts and the reproducibility of peak intensities For new sample types, test and define sample preparation (for example, initial sample processing, fractionation, or depletion), ProteinChip array selection, and array processing. For commonly used sample types (serum, plasma, CSF, and urine), protocols are provided in Appendix A Test and define SOPs for data acquisition, with emphasis on optimization of laser energy settings
Discovery	60 to 100s	For initial discovery of candidate biomarkers during this phase, apply methods that were defined during method development Use a sample number based on the expected complexity of the biological question relative to the heterogeneity of the study population. Complex clinical questions or analysis of samples from multiple collection sites may require larger sample sets If using controlled animal models (for example, inbred mice), use as few as 8–10 animals per group. Reduced numbers of animals or subjects may also be sufficient in longitudinal studies that are focused on change over time Use the maximum number of fractions and array chemistries allowed by the volume of samples available. Use a subset of possible procedures (including a subset of fractions and ProteinChip array chemistries) as an initial starting point only when absolutely required Refine data acquisition procedures as necessary
Validation	100s to 1,000s	Analyze samples from multiple collection sites and from other disease groups Include a subset of samples from the discovery phase to distinguish between failure of a marker to validate (population-dependent) and failure of the assay (inability to replicate results using discovery samples) Use the smaller, defined set of experiment conditions that yielded the biomarker candidates in the discovery studies
Clinical assay implementation	1,000s	Develop and apply chromatographic or antibody-based assays that are optimized to provide robust, sensitive, and quantitative protein biomarker assays

* Note that these phases differ somewhat from those of the general biomarker research process presented in Part I.

Select the Samples, Controls, and Standards

For any phase of biomarker research, select samples and controls using the general principles defined in Part II. For each phase of a project:

- Use statistical power calculations and the help of a biostatistician to select the number and types of samples (Table 4) that best address the goals of the phase and study
- Use sufficient numbers of sample replicates (at least 2, preferably 3)
- Select the types and number of controls to use (Table 5). Include one sample on each array for quality control (QC sample). The total number of QC sample spots can be reduced with very large sample sets
- For quantitation of specific proteins, use calibration and reference samples that are good models for the experimental samples. For example, dilute protein standards in a sample that is similar to that being analyzed (serum, plasma, CSF, etc.), ensuring calibrant concentrations are within the expected concentration range for the experimental samples

Table 5. Controls, standards, and reference samples used in ProteinChip SELDI analysis.

Sample	Description
Experimental Controls	
Healthy (normal) control	Samples from healthy individuals with no apparent diseases or disorders. Match these to the experimental disease group (for example, in terms of age, sex, ethnicity, location of sample collection, smoking status, diet, etc.)
Disease control	Samples from patients with other diseases or disorders that have clinical profiles mimicking those of the disease being studied. Match these to the experimental disease group
Treatment control	Samples from patients with the disease under investigation who did not receive a drug treatment (usually placebo or vehicle control). Use these for group comparisons in drug efficacy or drug toxicity studies
Time-course control	Samples from patients with the disease under investigation that were collected at an early time point, such as time zero (baseline) or pretreatment. Use these as a baseline for analyzing time-dependent changes in drug efficacy or drug toxicity studies
Calibration Standards	
Mass calibration standard	Molecular weight standards such as the ProteinChip all-in-one peptide standard or ProteinChip all-in-one protein standard II spotted on a single array and used to collect molecular weight (mass) calibration spectra. Conduct calibration daily or weekly, collect data at all laser energies used for optimization, and use data for recalibration of experimental data
Analyte concentration calibration standard (clinical assay phase)	Samples containing known proteins at defined concentrations. During the assay phase, use 4–6 different concentrations run in duplicate or triplicate on each bioprocessor. Use data from these standards to generate protein concentration vs. peak intensity curves for calculation of absolute sample concentrations
Reference Samples	
Quality control (QC)/ process control reference sample	Well-characterized pool of samples that is processed alongside experimental samples. Use spectra to: 1) monitor instrument performance during routine use, 2) compare protein profiles to historical reference sample profiles, and 3) calculate median coefficient of variation for peak intensities as a quantitative measure of reproducibility. Use a pool of study samples for laser energy optimization
Analyte reference samples (clinical assay phase)	Samples (often pooled clinical samples) containing known proteins at measured concentrations. Use these to: 1) calculate absolute concentration from the protein concentration vs. peak intensity standard curves, and 2) test the overall assay performance and accuracy of the standard curves

- If using a 96-well microplate for processing samples, use filtration plates for fractionation and the ProteinChip bioprocessor for binding of samples to arrays. The bioprocessor enables automated and reproducible profiling as it features wells capable of containing up to 150 µl of solution over each array spot. In addition, you may use the virtual notebook (VNB) feature of ProteinChip data manager software, Enterprise Edition, to randomize and track the sample layout for fractionation and spotting

Select Appropriate ProteinChip SELDI Array Chemistries

The choice of ProteinChip array chemistry depends on the proteome coverage required and whether the application requires general profiling or a specific protein assay. Different array types and wash conditions generate different protein profiles from the same sample. Combining these conditions yields a much broader picture of the proteome (Figure 2).

- For common biological fluids (serum, plasma, CSF, and urine), use the ProteinChip arrays and profiling buffers recommended in Appendix A
- For the general profiling needs of biomarker discovery, use a selection of array chemistries that provide different selectivities, or use different buffer stringencies for the arrays selected (Figure 2)
- For biomarker discovery using new sample types, test ProteinChip CM10, Q10, IMAC30, and H50 arrays. (ProteinChip NP20 arrays are not recommended for biomarker discovery due to their low selectivity.) Different buffer stringencies (such as different pHs for ion exchange surfaces, increasing organic or salt content for hydrophobic surfaces,

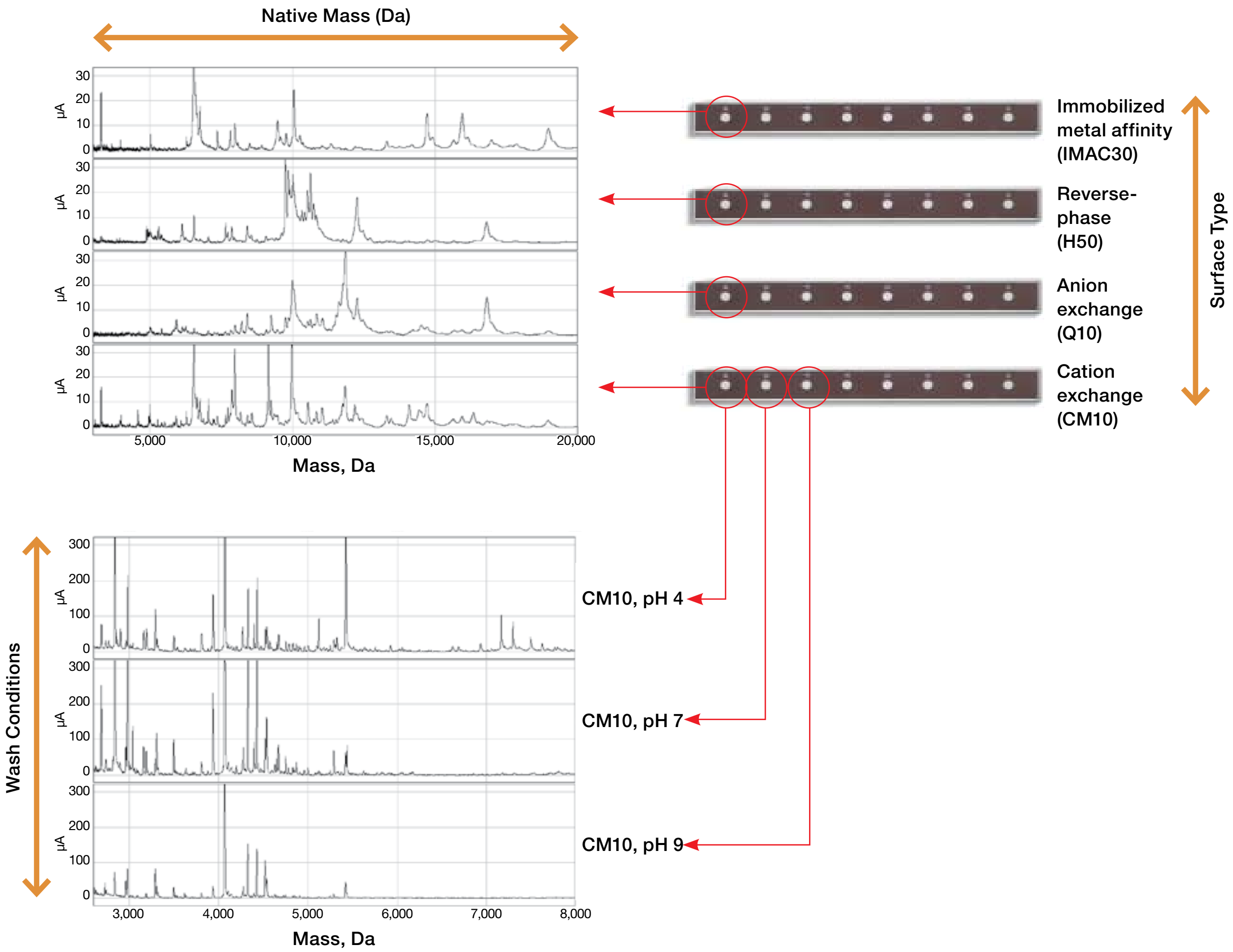


Fig. 2. Effects of different ProteinChip array surfaces and wash conditions. Choose combinations of ProteinChip array surface types and wash conditions to maximize the potential for protein biomarker discovery. Combining fractionation techniques with several array chemistries, different wash conditions, and mass measured by time-of-flight mass spectrometry dramatically increases proteome coverage.

or different metal ions for IMAC surfaces) can also be used to modulate the selectivity of these array chemistries (Figure 2)

- For targeted applications, such as validation or purification, select ProteinChip chromatographic arrays based on the experimental conditions that yielded the biomarker candidates during the discovery phase. Adjust buffer conditions as needed to optimize binding of the target analyte and reduce binding of other proteins
- For clinical assay implementation, redevelop optimized capture assay protocols using selected array chemistries and binding and wash conditions (as in validation studies), or use preactivated ProteinChip arrays for antibody immobilization

Select the Methods of Data Analysis

During assay design, plan the general approaches for downstream data analysis. Involve the expertise of a biostatistician to estimate the number of samples required for statistical power and to select analysis methods that minimize false discovery and over fitting; however, expect to refine these methods later, during data analysis.

Sample Preparation

Sample preparation is a potential source of analytical bias that is often overlooked and underemphasized. Sample preparation for a SELDI experiment involves sample processing (centrifugation, dilution, denaturation, etc.) often followed by fractionation or depletion methods. Throughout sample preparation, keep in mind that any liquid-handling step is a potential source of analytical variability.

Use Consistent and Appropriate Liquid-Handling Techniques

Proper and consistent laboratory practice is the primary consideration when selecting and planning liquid-handling techniques for biological samples. Use the following:

- Sufficiently large transfer volumes to ensure reproducible sample transfer. Transfer larger volumes first into an appropriate volume of dilution buffer before transferring smaller volumes of the diluted sample
- Protein-appropriate pipetting and transfer techniques (for example, use passivated surfaces, or pipet up and down from the bulk sample three times to condition pipet tips before transferring sample)
- SOPs for liquid handling that specify not only the techniques used, but also the equipment and consumables used for each step in the workflow
- Parallel processing of all samples; for example, process QC samples in parallel with experimental samples to monitor expected profiles and to calculate coefficients of variation (CVs)
- Automated liquid-handling systems (available from Beckman, Tecan, Hamilton, etc.) for increased throughput and to minimize analytical variability. Use calibrated multichannel pipets when automated systems are unavailable

Define Protocols for the Initial Processing of Samples

Depending on the sample type, some initial processing steps may be performed prior to binding onto ProteinChip arrays. For example, tissue samples are lysed using extraction buffers, and protease inhibitors, denaturants, and detergents are often added to samples before further processing. Since SELDI is compatible with a wide variety of detergents and salts, most lysis and storage buffers can be used (Table 6).

Other general recommendations for sample processing include the following:

- Ensure that the same protocols and buffers are used for all samples and, whenever possible, process all samples at the same time
- For cell or tissue lysates, include protease inhibitors to minimize artifacts generated by proteolysis (Table 6). Protease inhibitors are generally not required for serum or plasma samples
- Test effects of sample denaturation prior to analysis. Denatured proteins often give better results than native proteins for several fractionation techniques or for direct profiling. However, native samples may be required for specialized fractionation techniques, such as treatment with ProteoMiner™ beads
- Define a consistent and effective procedure for sample thawing (on ice, at 4°C, at room temperature, etc.) and aliquot samples into appropriately sized batches to avoid repeated freeze-thaw cycles. After thawing, vortex samples and centrifuge them briefly to remove particulates before processing
- When using tissue lysates, determine the total protein concentration of each sample and adjust all samples to the same concentration with extraction buffer before diluting the samples into binding buffer. This ensures that all samples are identical with respect to protein concentration (which affects peak number and intensity) and final buffer constituents (which affect binding stringency)

Table 6. Recommendations for use of common extraction and storage buffer components.*

Component	Recommendations
Detergents	Use nonionic detergents (CHAPS, OGP, Triton X, and NP40) in concentrations of up to 1% prior to dilution (1:10) in binding buffer. High detergent concentrations may inhibit binding to the hydrophobic ProteinChip arrays, but are compatible with all other arrays Avoid use of ionic detergents such as SDS, as they suppress ionization. In addition, detergents such as SDS make all proteins negatively charged and adversely affect binding to anionic and cationic arrays. If required for lysis, ionic detergents can be tolerated at low concentrations if arrays are washed thoroughly
Salts	If using high salt concentrations, dilute the sample or use ProteinChip arrays that are not sensitive to salt. High salt concentrations create more stringent binding conditions on anionic and cationic ProteinChip surfaces, but they do not interfere with binding to metal affinity or hydrophobic arrays
Denaturants	Use urea and guanidine at standard concentrations during sample preparation to reduce protein-protein interactions during fractionation and profiling, and to improve reproducibility and array selectivity. For binding to arrays, dilute samples (1:10) into binding buffer to reduce the denaturant concentrations
Reducing agents	Avoid using dithiothreitol (DTT), as it interferes with ProteinChip analysis; use weak (millimolar) solutions of β-mercaptoethanol instead
Protease inhibitors	Avoid peptide inhibitors (for example, aprotinin, leupeptin, and pepstatin), as they alter the protein profile. If using proprietary protease inhibitor cocktails, run buffer blanks on NP20 arrays and on each profiling array surface to detect any peptide inhibitors. Protease inhibitors are generally not required for serum and plasma
Anticoagulants (used to produce plasma samples)	Use citrate and EDTA, as these are generally preferred over heparin (the highly charged nature of heparin can significantly alter the chromatographic behavior of proteins and the resulting ProteinChip SELDI profiles). Though EDTA is compatible with most ProteinChip array surfaces, it may interfere with the direct analysis of samples on metal-coated IMAC surfaces, but this effect is reduced or eliminated by fractionation
Other	Do not use polyethylene glycol (PEG), as it is difficult to wash off and generates a strong, broad peak. Glycerol also interferes with detection. If either or both must be used, dilute the sample in binding buffer before application to an array to allow their removal in the washing step prior to matrix addition Avoid using diethylpyrocarbonate (DEPC, often used to inhibit RNase activity) for ProteinChip experiments, as it may covalently modify some proteins, leading to shifts in molecular weights, diminished signal intensities, or blank spectra. Autoclave DEPC-containing solutions before use to remove residual DEPC, and do not use DEPC-treated consumables such as pipet tips and microcentrifuge tubes

* For more details about chemicals that interfere with protein analysis, refer to the ProteinChip SELDI System Applications Guide, Volume 1.

- For more recommendations on serum and plasma sample processing for SELDI analysis, refer to the ProteinChip SELDI System Applications Guide, Volume 2

Use Fractionation and Depletion Techniques to Improve Detection of Low-Abundance Proteins

Different sample types often differ dramatically in the amount and diversity of protein species they contain. For example, urine is usually quite dilute and may require some form of protein concentration or serial incubation with ProteinChip arrays to maximize protein binding to the array surface. In contrast, serum and plasma are highly complex sample types, presenting a large dynamic range of protein concentrations, with the 20 most abundant (and known) proteins representing 97–99% of the total protein mass. Though addition of any sample-handling step increases chances for variability and analytical bias, depletion and fractionation of complex sample types such as serum and plasma prior to SELDI analysis increases peak counts and improves detection of low-abundance proteins.

Fractionation, the separation of protein mixtures into discrete fractions according to their physicochemical properties, reduces sample complexity for enhanced detection of low-abundance proteins. A variety of approaches are used for sample preparation in proteomics workflows, including chromatographic,

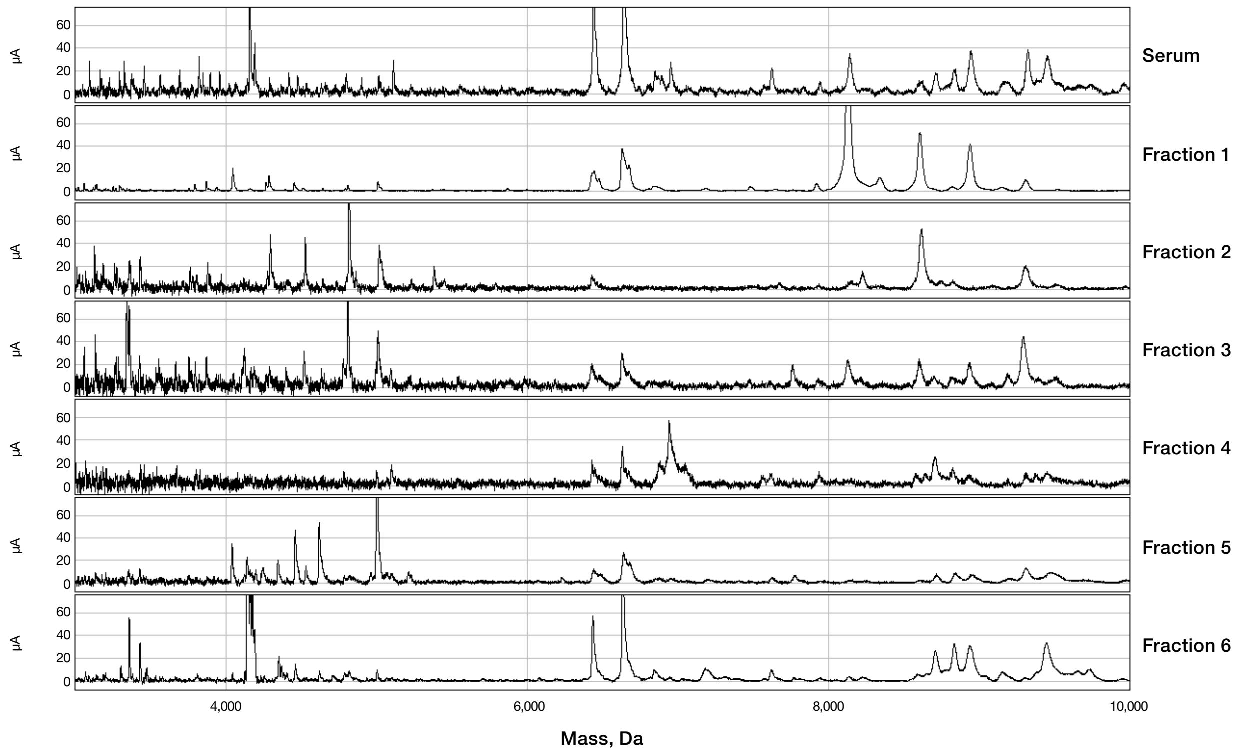


Fig. 3. Fractionation of serum samples by strong anion exchange chromatography. Human serum was fractionated using the ProteinChip serum fractionation kit to yield six fractions. Each of the fractions was analyzed, along with the unfractionated serum, using ProteinChip CM10 arrays. Shown is the 3–10 kD mass range for the six fractions compared to untreated serum. Note that the spectra from each of the fractions reveal peaks that might otherwise go undetected or that are present at very weak intensity in the spectrum of the unfractionated sample.

reagent-based, and electrophoretic methods. These methods separate proteins by a wide range of physical properties including solubility, hydrophobicity, subcellular location, size, or isoelectric point (Bio-Rad bulletin 3145). An effective approach recommended for fractionation of serum, plasma, or other complex biological fluids upstream of SELDI analysis uses strong anion exchange chromatography to separate proteins into six fractions, each of which may then be applied to three or four different chromatographic array chemistries (Figure 3). This fractionation method is commercially available as the ProteinChip serum fractionation kit (catalog #K10-00007). It requires only 20 µl of mammalian serum or plasma per sample, is amenable to automation, and reproducibly increases the number of proteins that can be observed in SELDI analyses.

Depletion helps improve detection of low-abundance proteins by selectively removing high-abundance proteins from samples. Immunodepletion is a popular depletion approach that uses antibodies, which are often bound to chromatographic supports, to selectively remove proteins from samples. Immunodepletion is commonly used to remove high-abundance proteins, such as albumin and immunoglobulins from serum and plasma samples. Though it is often effective, it suffers from several drawbacks: the cost of the antibodies, the resulting sample dilution, and the fact that any depletion strategy may be associated with the loss of potential biomarkers — some peptides and proteins of interest might either bind to the depleted proteins or be disease-associated proteolytic fragments of the depleted proteins — which may be recovered by elution of the capture resin.

As an alternative to immunodepletion, Bio-Rad has developed the ProteoMiner technology (see sidebar, page 34) to enable the simultaneous removal of excess high-abundance proteins and enrichment of low-abundance proteins. This method uses short peptide fragments (hexapeptides) as affinity recognition sites (Castagna et al. 2005, Fortis et al. 2006a and 2006b, Righetti et al. 2006, Righetti and Boschetti 2007).

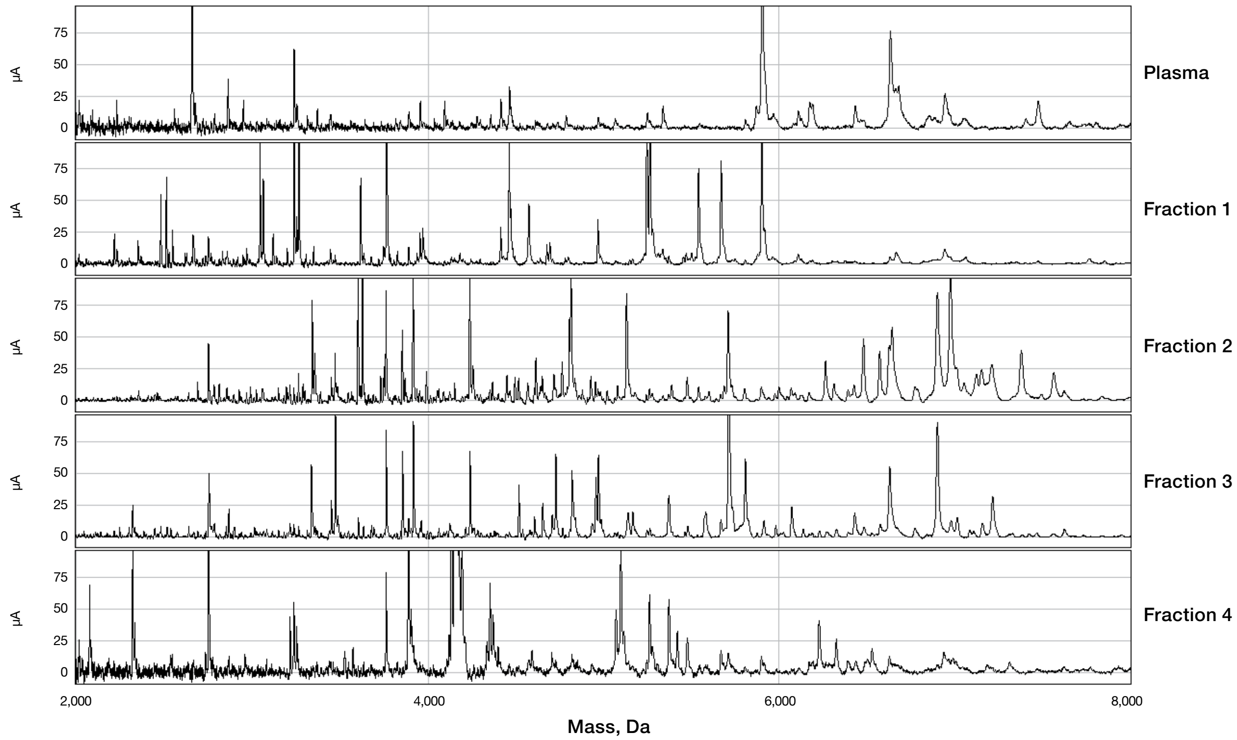


Fig. 4. Fractionation of plasma samples by ProteoMiner technology. Human plasma was fractionated using ProteoMiner technology to generate four fractions. Each of the fractions was analyzed, along with the untreated plasma, using ProteinChip IMAC30 arrays. Shown is the 2–8 kD mass range for the four fractions compared to untreated plasma. The flowthrough fraction is generally analyzed along with the four fractions, and its spectrum is very similar to that of untreated sample (not shown). Simultaneous depletion of high-abundance proteins and enrichment of low-abundance proteins greatly increases sensitivity of detection, with an increase in total peaks detected relative to untreated plasma or strong anion exchange fractionation (data not shown).

Use of this approach can dramatically reduce the dynamic range of protein concentration in samples like serum and plasma. This results in improved SELDI peak counts and enhanced detection sensitivity (Figure 4). ProteoMiner technology has been used successfully with a number of other sample types (cell lysates, urine, and lavage) in addition to serum and plasma.

To maintain reproducibility during fractionation or depletion, pay careful attention to the standard precautions defined in other sections in addition to the following:

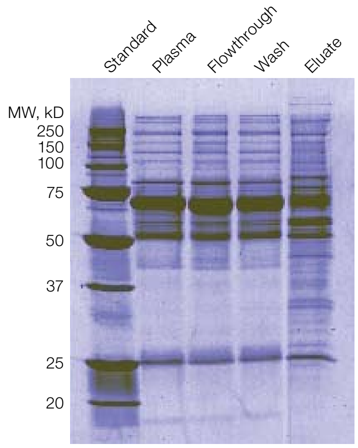
- Treat as many samples at a time as possible, including patient controls and QC samples, to minimize analytical bias. To improve throughput and reproducibility, use 96-well microplate formats for sample preparation
- Use the same lot or batch of chromatographic resin, premixed buffers, and other reagents for all samples and controls
- Randomize the order in which samples are prepared or treated. If running multiple 96-well microplates, for example, randomize samples across all plates to reduce bias. However, if analyzing multiple time points or dosages from the same individual, analyze all samples from the same patient on the same microplate in order to minimize the effect of plate-to-plate bias in paired statistical analyses (for example, Wilcoxon or repeated measures tests)
- When using a 96-well microplate format for sample preparation, use the VNB feature of ProteinChip data manager software, Enterprise Edition, to determine the layout of samples and controls on the microplates and subsequent arrays. The VNB not only allows tracking of samples throughout preparation and analysis, it is capable of generating a randomized layout of samples for sample preparation and analysis
- Process or freeze collected fractions as soon as possible after fractionation. If fractions are frozen, minimize freeze-thaw cycles by aliquoting into multiple plates if processing will be spread over several days (for example, different array types on different days)

ProteoMiner Technology

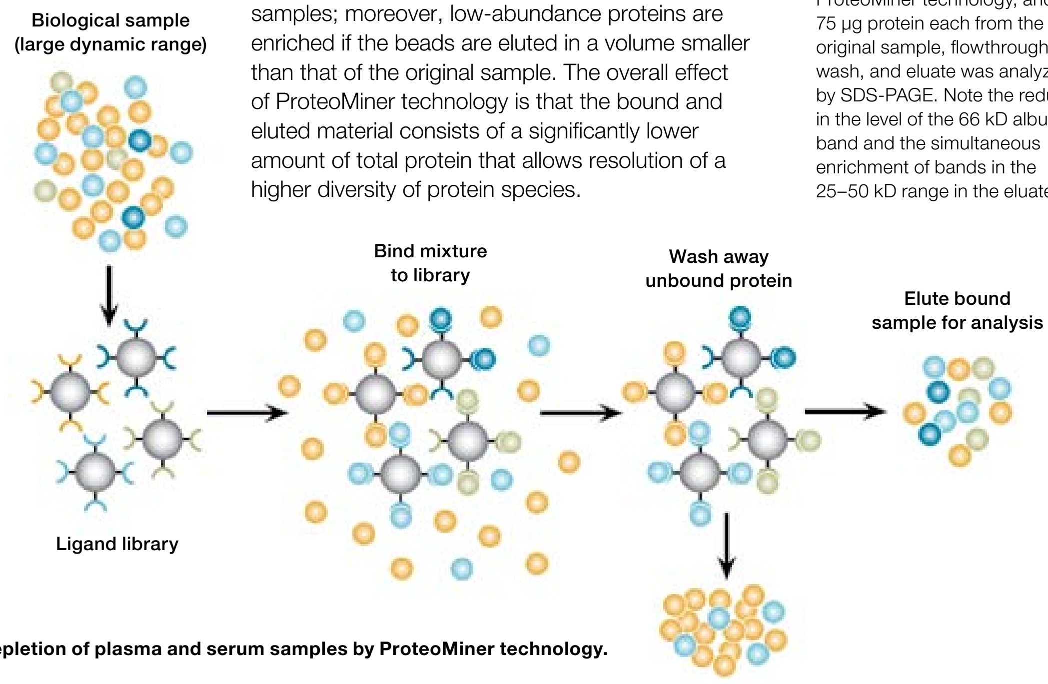
ProteoMiner protein enrichment technology is a sample preparation tool used to reduce the dynamic range of proteomes from complex biological samples such as serum and plasma. It provides a depletion strategy that overcomes most, if not all, of the disadvantages of immunodepletion while simultaneously reducing the concentration of high-abundance proteins and enriching low-abundance proteins.

This technology employs a combinatorial library of hexapeptides bound to a chromatographic support. Combinatorial synthesis allows the creation of a large library of unique hexapeptides, with each hexapeptide bound to a stationary support, or bead. Each bead featuring a unique ligand is expected to bind specifically to one or a small number of different proteins in a mixture, and the library of all possible sequences binds proteins up to the capacity of available beads.

When a complex biological sample is applied to the beads, high-abundance proteins saturate their ligands, and excess proteins are washed away. In contrast, low-abundance proteins do not saturate their binding sites and thus, different samples retain similar relative expression levels to the original samples; moreover, low-abundance proteins are enriched if the beads are eluted in a volume smaller than that of the original sample. The overall effect of ProteoMiner technology is that the bound and eluted material consists of a significantly lower amount of total protein that allows resolution of a higher diversity of protein species.



SDS-PAGE analysis of plasma treated with ProteoMiner technology. A plasma sample (1 ml) was treated with the ProteoMiner technology, and 75 µg protein each from the original sample, flowthrough, wash, and eluate was analyzed by SDS-PAGE. Note the reduction in the level of the 66 kD albumin band and the simultaneous enrichment of bands in the 25–50 kD range in the eluate.



Depletion of plasma and serum samples by ProteoMiner technology.

Array Processing

Array processing involves the steps used to pretreat, bind samples to, and wash ProteinChip arrays prior to analysis. The basic steps of array preparation are the same for all sample types; however, these steps may require optimization of parameters such as sample dilution, buffer composition, and binding times. Follow the general protocols for array processing described in the ProteinChip SELDI System Applications Guide, Volume 2 and in Appendix A of this guide.

The following general recommendations for array processing apply to any phase of a proteomics study and to several of the array preparation steps:

- During array processing, wear gloves to reduce the risk of contamination. Use powder-free, nonlatex gloves (such as nitrile), as latex generates peaks that complicate data interpretation. Do not touch the spots on the arrays
- Within a single condition (fraction/array type/matrix combination), use a single lot of arrays, buffers, and matrix
- Use the same buffer composition for array equilibration, sample binding, and washing
- Use a ProteinChip bioprocessor, appropriate microplate mixer (for example, a MicroMix 5 mixer, Diagnostic Products Corp.), and consistent incubation times and temperatures for sample binding
- Monitor bioprocessors for bubbles, and take precautions to minimize bubble formation (refer to the ProteinChip SELDI System Applications Guide, Volume 2). Bubbles at the bottom of a bioprocessor well over the ProteinChip array spot prevent buffers and samples from reaching the array surface; this may lead to blank spectra and poor reproducibility. One option for preventing or removing bubbles is to centrifuge the bioprocessor for 1 minute at 250 x g to ensure the buffer is directly in contact with the array surface

Determine the Sample Layout for Each Array

If using a 96-well microplate format for fractionation, use the same layout of samples and controls for fractionation and spotting. Use the VNB to plan and randomize the sample layout on each ProteinChip array prior to fractionation, as described previously. After fractionation, complete array preparation and data collection on one fraction/array chemistry/matrix condition before continuing to the next condition.

When binding unfractionated samples directly onto ProteinChip arrays, follow these recommendations, which are similar to those described for sample fractionation using a 96-well microplate format:

- Use the VNB to track all available clinical information and sample preparation details, and to associate array bar codes with data collection parameters
- Process as many samples at a time as possible to minimize analytical bias. Use ProteinChip bioprocessors and automated liquid-handling systems to minimize variability and increase throughput
- Randomize the sample layout on each bioprocessor. If running multiple bioprocessors, randomize samples across all bioprocessors to reduce bias. However, if analyzing multiple time points or dosages from the same individual, analyze all samples from the same patient on the same bioprocessor to minimize the effect of plate-to-plate bias in paired statistical analyses (for example, Wilcoxon or repeated measures tests)
- Include appropriate control and calibrant samples throughout the bioprocessor layout. In addition, include occasional blank wells (use only buffer during sample incubation) to assess contamination and well-to-well sample carryover. Also, use pools of study samples for laser energy optimization

Optimize Sample Dilution and Buffer Composition

Sample dilution and buffer composition are two of the most important parameters affecting the quality of a SELDI profiling experiment. Appendix A of this guide provides recommendations for appropriate sample dilutions and buffers for use with serum, plasma, CSF, or urine samples; however, further optimization may be required for other sample types or specific applications.

Loading too much or too little sample onto an array affects spectral quality by generating either saturated or undetectable peaks, respectively. As in any chromatography experiment, overloading the substrate leads to nonspecific binding. In either case, the reproducibility of peak intensity is compromised and the total peak count is reduced (Figure 5).

The composition of the buffer used for ProteinChip array equilibration, sample binding, and postbinding washing can also impact protein binding to an array and, therefore, the resulting spectra (Figure 6).

Optimize protocols for diluting sample into binding buffer to maximize both the peak counts and the reproducibility of peak intensities:

- Simultaneously examine the effects of different buffer compositions along with different sample dilutions. On a set of arrays, prepare a series of different buffer and dilution combinations for analysis
- To maximize the number of proteins detected, use both a high- and a low-stringency buffer. This approach is particularly useful with unfractionated samples
- Analyze a dilution series (for example, within the range of 50–2,000 µg/ml total protein) using duplicate samples and identical buffer solutions. Select the dilution and buffer stringency that yield the most reproducible profiles and highest peak counts
- Use fractionation or protein depletion methods to decrease sample complexity before SELDI analysis. This is particularly useful if there is a wide range of protein concentrations in the sample, and if some peaks disappear and others become saturated in the dilution series
- Optimize buffer composition and sample dilution for each different ProteinChip array type used
- Maintain consistent sample dilution and buffer composition for future studies using similar samples

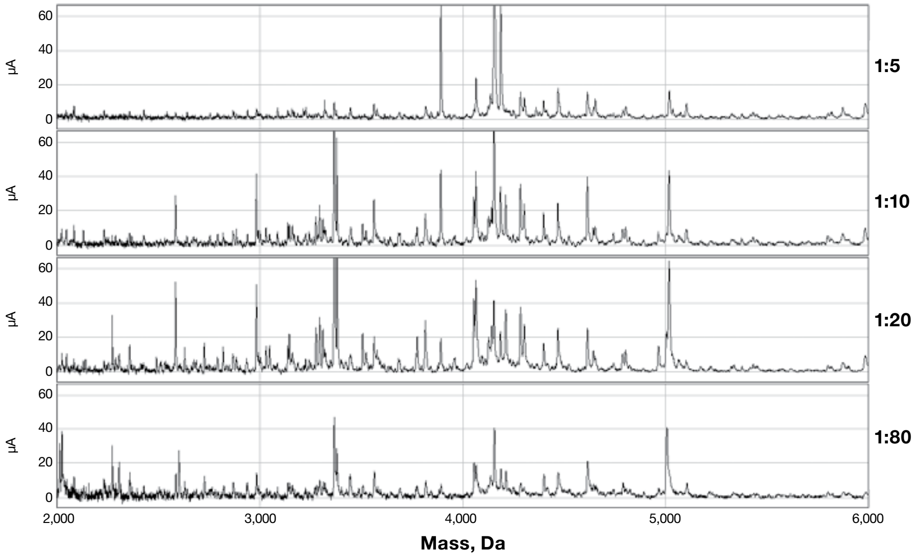


Fig. 5. Effect of sample dilution on total peak count. Human serum diluted in pH 7 buffer was bound to CM10 arrays. At high protein concentrations (1:5 dilution), the surface was likely saturated by high-abundance proteins, reducing binding and detection of low-abundance species and resulting in low peak counts. In this example, total peak count reached an optimum at 1:20 dilution; peak counts were lower at the lower and higher dilutions.

Standardize the Methods of Matrix Application

An essential component of a ProteinChip SELDI experiment, the matrix assists in desorption and ionization of analytes. The matrix is applied in organic solvent, and it solubilizes many of the proteins on the array surface. As the matrix solution dries, it cocrystallizes with the proteins. These crystals absorb the laser energy and generate the ionized proteins detected by the ProteinChip SELDI instrument.

The two matrix molecules most commonly utilized for SELDI protein profiling analyses are sinapinic acid (SPA, catalog #C30-00002) and α-cyano-4-hydroxycinnamic acid (CHCA, catalog #C30-00001):

- Use SPA for general analysis of peptides and proteins
- Use CHCA for enhanced detection of low-mass peptides
- For maximum proteome coverage, prepare duplicate sets of arrays, and add SPA to one set and CHCA to the other set. Do not add both SPA and CHCA on the same array
- Use other matrix molecules if they are more suitable for specific proteins or applications. For example, matrices that enable efficient laser desorption and ionization of glycosylated proteins include 2,5-dihydroxybenzoic acid and EAM-1 (proprietary formulation, catalog #C30-00003)

The quality and chemical nature of the matrix used have a dramatic effect on the quality and reproducibility of data, as does the method of matrix application. Follow the recommendations given in the ProteinChip SELDI System Applications Guide, Volume 2 as well as the following:

- Use fresh reagents for preparing matrix solutions
- Use consistent timing for both matrix applications. Apply the matrix shortly after the water rinse dries (20–30 minutes after the water rinse), and again shortly after the first matrix addition dries (10–20 minutes). Consistent timing between the two matrix applications is more important than any precise time for drying; however, the drying time between applications should not exceed 30 minutes. Longer drying times significantly reduce the signal intensity and assay reproducibility
- Maintain consistent temperature and humidity during matrix application and array drying
- Use a well-calibrated, small-volume repeat pipet and consistent liquid-transfer procedures for matrix addition
- Consider using a dedicated noncontact, fixed-tip, nanovolume liquid-dispensing instrument to improve reproducibility and throughput (for example, those offered by BioDot, Inc. are compatible with ProteinChip bioprocessors)

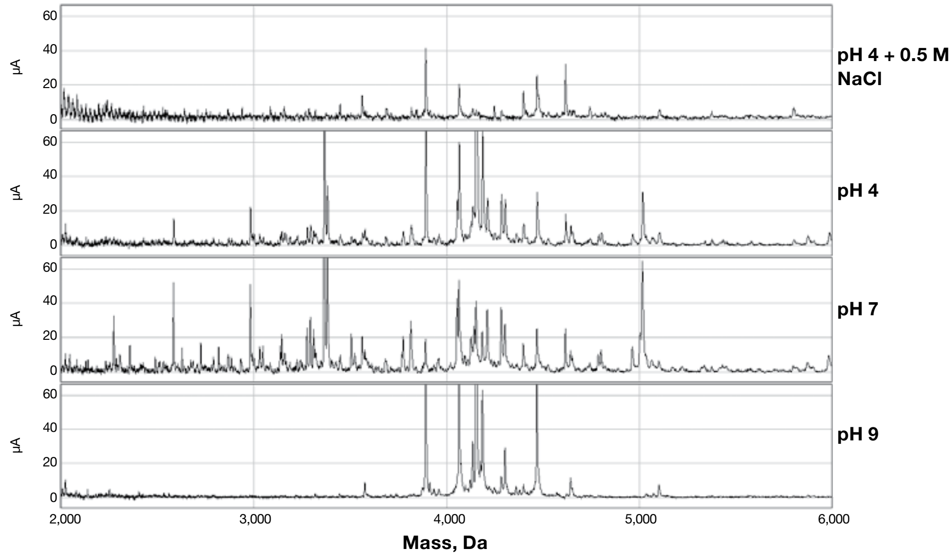


Fig. 6. Effect of binding buffer on total peak count. Human serum diluted 1:20 in various buffers was bound to CM10 arrays. At low stringency (pH 4), the surface was likely saturated by high-abundance proteins, reducing binding and detection of low-abundance species. At high stringencies (pH 9 and pH 4 + 0.5 M NaCl), only proteins with the highest affinities bound to the surface, reducing profile complexity. In this example, total peak count was maximized by using pH 7 binding buffer.

Data Collection

Data collection with the ProteinChip SELDI system requires optimization of instrument settings and implementation of SOPs. Follow the procedures for operating the ProteinChip SELDI system and collecting data described in the ProteinChip Data Manager Software Operation Manual, chapters 5 and 7.

Perform Regular Instrument Maintenance and Calibration

Qualification and calibration of analytical instruments is a formal process of testing and documenting that an instrument is fit for its intended use and that it is maintained and appropriately calibrated. ProteinChip SELDI system qualification and calibration kits ensure optimum performance of the ProteinChip SELDI instrument. These kits improve the reproducibility and reliability of SELDI data and use the same products and techniques that are used for testing by Bio-Rad quality control and system service teams. Routine performance of these procedures minimizes potential sources of analytical variability related to instrument function. In addition, monitoring and calibration require routine performance qualification using appropriate reference and calibration samples selected for the specific SELDI study.

- Use the ProteinChip detector calibration kit (catalog #C70-00082) to automatically adjust the detector voltage and improve reproducibility over time. These adjustments help maintain a constant gain, stabilizing the output over the detector's lifetime
- Use the ProteinChip system check kit (catalog #C70-00081) to collect data and generate reports that show reader compliance to a set of four operational specifications for sensitivity and resolution
- Use the ProteinChip OQ kit (catalog #C70-00080) in regulated laboratories to collect data and generate reports that show reader compliance to nine operational specifications for sensitivity, resolution, and mass accuracy
- Include reference samples (Table 5) throughout the study to monitor instrument performance during routine use. Compare protein profiles to historical reference sample profiles and calculate median CVs for peak intensities
- Collect data for mass calibration standards (Table 5) weekly at each of the laser energy settings used for optimization of acquisition protocols (see next section). Use these data to generate a calibration equation, and use this equation later to calibrate experimental sample spectra

Optimize the Acquisition Protocols

The ProteinChip SELDI instrument uses acquisition protocols to acquire data from a spot or portion of a spot on a ProteinChip array. The ProteinChip instrument is programmed with a set of acquisition parameters for peptides and proteins that can be combined with optimized laser energy settings to create data acquisition protocols. Table 7 summarizes the various acquisition parameters that can be optimized, as well as recommended starting points for general profiling applications.

- Test the acquisition protocols using the recommended settings in Table 7 on a pooled sample before collecting data from study samples. A pool of the experimental samples is preferred over a reference sample for optimization since it more accurately reflects the experimental samples
- To ensure sufficient coverage of the entire mass range, separately optimize the acquisition settings for both the low-mass (0–20,000 Da) and high-mass (>20,000 Da) proteins using several partitions on 3–4 pooled sample spots. Note that more partitions (smaller percentages of the spot surface) may be used during optimization than are recommended for data collection on experimental samples

Table 7. Starting SELDI parameters for spectral acquisition during general protein profiling.*

Parameter	Description	Recommended Settings		
		Low-Mass Proteins		High-Mass Proteins
Acquisition Settings		CHCA	SPA	SPA
Acquisition mode	Use these default settings for general protein profiling applications	Source at 25 kV; positive ions		Source at 25 kV; positive ions
Mass range (Da)	Mass collection range. Affects the amount of data collected, but does not affect the peaks themselves	0–25,000		0–200,000
Focus mass (Da)	Mass around which optimal peak resolution is obtained	5,000		19,000
Matrix attenuation (Da)	Detector blanking portion of the spectrum for which the detector is protected from matrix saturation. Use higher matrix attenuation values when using higher laser energies	1,000		5,000
Sampling rate (MHz)	For better resolution at lower masses, use higher sampling rates, which generate larger data files. Use lower sampling rates for high-mass data acquisition, where peaks have lower maximum resolution	800		400
Calibration	Calibration equation generated using mass calibration standards and for use during acquisition of new spectra or recalibration of existing spectra	Equation generated using ProteinChip all-in-one peptide standards		Equation generated using ProteinChip all-in-one protein standards
Shot Sequence Settings				
Acquisition method	Specifies whether the laser energy is maintained constant (SELDI Quantitation) or adjusted (Auto Laser Adjust) during data acquisition	SELDI Quantitation	SELDI Quantitation	SELDI Quantitation
Warming shots (nJ)	High laser energy often used prior to collection of data transients with the lower laser energy setting. Use 2–4 warming shots per pixel. Do not add warming shot data transients with the data averages	Data shot energy +200 nJ	Data shot energy +250 nJ	Data shot energy +500 nJ
Data shots (nJ)	Laser energies used for collection of data transients and included into data averages. Requires optimization prior to data acquisition. Use 5–25 data shots per pixel	800–2,200 (step by 200 nJ during optimization)	2,000–3,500 (step by 250 nJ during optimization)	3,000–8,000 (step by 500 nJ during optimization)

* Acquisition and shot sequence settings refer to basic protocol parameters as displayed in the data collection protocol wizard in ProteinChip data manager software. Use these parameters for general profiling applications and as starting points for optimization of specific analyte assays. Note that spectra collected from arrays prepared with different matrix molecules emphasize different classes of proteins and require use of different instrument settings for data acquisition.

- If different matrix molecules were used for sample preparation, optimize each separately for the relevant mass range
- After optimizing laser energy settings, keep all settings constant during collection of a data set for all samples associated with that condition (fraction/array type/ matrix combination)

Of the data acquisition parameters listed in Table 7, laser energy often has the most significant effect on spectra (along with sample dilution and binding buffer, as previously described) and should be optimized first. Different protein peaks exhibit optimal intensities at different laser energies, and laser energies that are too high or too low yield saturated or undetectable peaks, respectively. Insufficient laser energy causes insufficient extraction of ions from the array surface, resulting in a low peak count. Excessive laser energy yields off-scale readings, unstable baselines, and broad, flat-topped peaks that make accurate quantitation of expression differences impossible. With the optimum laser energy, peaks are sharp and well-resolved. General starting conditions for a laser energy study, which are defined in Table 7, depend on the matrix used and the mass range of interest.

Additional acquisition parameters that may require optimization, especially when assaying for a small number of well-defined analytes, include detector blanking (matrix attenuation) setting, focus mass, sampling rate, and number of shots to acquire into the spectrum average.

Acquire Data Using Default Processing Parameters

As spectra are acquired, they are stored in folders in ProteinChip data manager software. Processing includes baseline subtraction, filtering, noise estimation, and default mass calibration. If the default processing parameters are not appropriate, spectra can be reprocessed later during data analysis (see Data Analysis).

- Complete data collection on one experimental condition (fraction/array chemistry/matrix combination) before continuing to the next experimental condition
- Acquire data for arrays prepared with SPA as the matrix at two settings to optimize for low- and high-mass (or mass-to-charge ratio, m/z) proteins
- Acquire data for arrays prepared with CHCA as the matrix at one setting optimized for the low-mass range
- Collect data using 1 of 3 partitions for low laser energy readings using SPA or CHCA. For high laser energy readings using SPA, use partition 2 of 3. Default settings are 10 laser shots (transients) per pixel collected and averaged per spectrum after 2 warming shots per pixel, which are not included in the averages. Collect more than 10 transients per pixel (up to a maximum of 25) when necessary to reduce baseline noise
- If collecting data at different times, especially across days, weeks, or months, include reference samples (Table 5) for analysis of relative peak intensities instead of absolute peak intensities. Reference samples must contain the peak(s) of interest for proper data normalization, so use pooled samples

Data Analysis

Recommendations for the annotation, grouping, and processing of ProteinChip SELDI spectra are described here to present the overall data analysis workflow. For steps and procedures not covered in detail here, refer to the appropriate sections in the ProteinChip Data Manager Software Operation Manual. Since the details of statistical analysis depend on the nature of each individual study, only general guidelines and key considerations for statistical analyses are covered here.

Ensure Proper Annotation of Spectra

Annotations are used for grouping spectra and as the basis for grouping sample cohorts during determination of biomarker candidates and development of classification algorithms. Ensure the accuracy and completeness of the annotation associated with each spectrum, and update this information as necessary. Include all sample and patient information as well as details of how the samples and arrays were processed.

When using ProteinChip data manager software, Enterprise Edition, enter all annotations in the VNB. Annotations entered in the VNB before spectra are collected are automatically saved as metadata along with the spectral data. If annotations are entered or updated after spectra are collected, synchronization with the VNB will be required to update the metadata.

Process Spectral Data

Spectrum processing involves performing mass calibration, baseline subtraction, and normalization of the spectra and often requires some additional optimization of the default settings used during data collection. Parameters vary for different conditions. Optimization is best approached by observing the effects of adjusting parameters on a few representative spectra. The largest differences in parameters exist for data acquired using different matrix molecules and mass optimization ranges.

Mass Calibration

The time-of-flight data for peptides derived by mass spectrometry are converted to molecular weight data by mass calibration. Data collected using known peptides or proteins are used to derive a calibration equation, which is then applied to the spectra of experimental samples. This mass calibration procedure is analogous to using molecular weight protein standards in gel electrophoresis. For best results:

- Collect mass calibration spectra on a weekly basis using the ProteinChip all-in-one peptide standard (catalog #C10-00005) or ProteinChip all-in-one protein standard II (catalog #C10-00007)
- To ensure accuracy, use calibrants that span the mass range of interest and prepare calibrants separately for each matrix molecule used
- Collect data for all matrix molecules and laser energies that will be used during optimization. Use the same set of data collection protocols that will be used for the experimental samples
- Externally calibrate spectra using a calibration equation collected using all of the same parameters (laser energy, focus mass, etc.) used for data collection

Baseline Subtraction

More accurate measurements of peak heights (intensities) are obtained by first adjusting the spectral baseline, which may be artificially elevated by the matrix desorption process. Adjust parameters so that the calculated baseline follows the baseline contours of the spectrum. Ensure that the baseline does not alter peak areas or intensities and that it is applicable to all spectra. During baseline subtraction:

- When evaluating baseline parameters, turn off baseline subtraction and check the fit of the calculated baseline to the data. Note that all peak intensities calculated and exported by ProteinChip data manager software include baseline subtraction using the parameters stored at the time of peak intensity calculation

- Set the noise estimation range to include the mass range of interest and exclude data below the blanking (attenuation) value used for data collection

As a starting point, process spectra using the default analysis parameters with the following changes, which are general guidelines that may need optimization for each data set:

- For spectra optimized for the low-mass range, subtract the baseline using an empirically determined setting (generally in the range of 10–20 times the expected peak width). Adjust the noise definition by modifying the segment for measuring local noise to include the mass range from approximately 2,000 Da (select this setting to be at least 500 Da above the detector blanking setting) to the end of the spectrum
- For spectra optimized for the high-mass range, make the adjustments recommended for low-mass range optimization and the following: 1) set baseline smoothing to approximately 10 points, and 2) set peak filtering to approximately 1 times the expected peak width and the segment for measuring local noise to the mass range from approximately 10,000 Da (select to be at least 1,000 Da above the detector blanking setting) to the end of the spectrum

Normalization

The normalization process calculates an average total ion current (TIC) from all the spectra and adjusts the intensity scales for all the spectra to this average. Once a spectrum is normalized, the resulting normalized peaks intensities are used in downstream analysis, such as biomarker analysis. The TIC normalization method works best with samples that have similar protein profiles. TIC normalization assumes that the contribution to the TIC of peaks that change based on treatment or disease is insignificant compared to the overall protein profiles. Should the TIC (or total protein content) differ significantly between relevant groups, TIC normalization may not be appropriate. This has not been observed with human serum or plasma samples, but can be an issue with other sample types and model systems, and in diagnostic assay applications, where a few analytes of interest significantly affect the TIC. In these cases, use other normalization procedures (for example, use the peak intensity ratio to an internal spike with known proteins or peptides) and adjust the total protein content, if needed, prior to sample processing. For normalization:

- Use the TIC method to linearly scale the intensities of a set of spectra to account for spectrum-to-spectrum variations due to ionization differences. By default, ProteinChip data manager software calculates an average normalization coefficient, but for convenience, an external normalization coefficient of 1.0 is often used
- Adjust the starting mass for normalization for different laser energies and detector blanking settings to be equal to or greater than the lower cutoff for noise definition. Start from at least 2,000 Da (spectra optimized for low mass) and 10,000 Da (high mass) to ensure that the chemical noise in the low-mass range is excluded from normalization. In most studies, spectra are normalized for m/z of 2,500–25,000 (spectra optimized for low mass) and for m/z of 10,000–200,000 (high mass)
- Evaluate spectra with extreme normalization factors as a quality control measure to identify and eliminate spectra of poor quality prior to statistical analysis (see Evaluate the Quality of the Data)
- After optimization, use the optimized settings and workflow to process all spectra within a folder

Group Spectra Into Folders

Organize spectra using ProteinChip data manager software into separate folders within a project folder. Base the organization on differences in experimental conditions or other variables (for example, fraction number, array chemistry, stringency of binding or wash buffers, matrix or mass optimization range, and laser energy used for data acquisition). This organization enables independent analysis of data sets with identical histories of sample treatment, array processing, and data collection.

First, group experimental spectra into separate data folders according to experimental conditions. As an example, six anion exchange fractions separated on three array chemistries (with one stringency wash per chemistry), using SPA as the matrix, and with data acquired at two laser energies per spot should be grouped into 36 separate analysis folders: (6 fractions) x (3 array chemistries) x (1 matrix) x (2 laser energies).

Second, separate spectra into folders for peak detection and analysis. Group spectra from QC or process control samples separately from spectra of experimental samples. Use data from the QC or process control samples to calculate median or pooled CVs of peak intensities and to monitor the quality and reproducibility of SELDI experiments.

Evaluate the Quality of the Data

The next step in data analysis is to evaluate the quality of the data, eliminate any spectra of poor quality, and evaluate the overall quality of the assay before proceeding further into statistical analysis.

One objective method for identifying and eliminating spectra of poor quality uses normalization factors to select spectra for exclusion. To use this method, export the normalization factors for each set of spectra into Excel software and calculate the median value. Remove any spectra that have a normalization factor greater than twice the median value. Failing spectra are usually caused by air bubbles in the bioprocessor or particulate matter in the sample. If all replicates fail to meet these criteria, there may be a problem with the integrity of the sample; in such cases, remove the samples from further analysis.

To monitor the quality of the assay, process and monitor the spectra generated by reference samples in a manner identical to that used with experimental samples. Apply the reference sample to at least one spot on each array. The number of reference serum spots can be reduced for larger studies. Use the peak detection settings that are used with the experimental samples for the reference samples, with the following changes: detect peaks in 50% of spectra and turn off the second pass of peak detection. As with experimental samples, these peak detection parameters may require optimization.

Export the peak information — including mass and peak intensity — for each peak to Excel software. Calculate the median CV for each peak and the pooled or median CV of all peaks. Expect pooled or median CVs in the range of 10–20% (direct profiling without fractionation) or 15–25% (profiling after fractionation). Significantly higher median CV values indicate analytical bias and should be investigated, as they may indicate that the study should be repeated.

Detect, Label, and Cluster Peaks Within One Condition

The cluster wizard feature of ProteinChip data manager software performs automatic peak detection, labeling, and clustering of peaks with user-defined settings. Performing an analysis for potential biomarkers requires the creation of clusters from the peaks in the spectra. A cluster in ProteinChip data manager software is defined as a group of peaks of similar mass that are treated as the same substance across multiple spectra. Once the peaks are clustered, use the peak intensities from each spectrum to statistically compare relative expression levels for that substance between different sample cohorts.

Perform detection, labeling, and clustering of peaks for spectra from samples that were processed and analyzed under identical conditions. Prior to clustering, process the spectral data for these samples uniformly and group them as described previously. Use consistent settings selected to meet the desired sensitivity of peak detection. When optimizing peak detection settings, the objective is to accurately label all robust peaks without labeling spurious peaks generated by chemical or electronic noise. However, some overlabeling can be tolerated, since these peaks usually fail to demonstrate any statistical significance and would be excluded during quality control of candidate markers. As general guidelines (developed for fractionated serum or plasma), use the following settings as starting points from which to optimize:

- Use settings of 5 times the signal-to-noise ratio and a valley depth of 3 for the first pass of peak detection. A second detection pass may also be included
- For experimental samples, select for a peak cluster to be created if a given peak is found in at least 10–20% of the spectra, depending on the number of samples and different groups used in the study
- Define the mass window for peak clustering to approximately 0.3% of the peak mass (low-mass range) and to approximately 2% of the peak mass (high-mass range). The larger mass window for the high-mass proteins accounts for broadening of peaks due to posttranslational modifications
- For validation or assay phases, manually label peaks of interest and set the software to cluster only user-detected peaks in order to ensure labeling of all relevant peaks. Alternatively, save relevant peak clusters and automatically apply them for peak detection

The result of this process is a set of peak clusters with associated intensities for each sample. Use these peak cluster data to determine biomarker candidates using statistical tools within ProteinChip data manager software. Alternatively, export the data to a Microsoft Excel spreadsheet for analysis using ProteinChip pattern analysis software or other external software packages.

Perform Univariate Statistical Analyses

Statistical tests are generally used to screen for peaks that show significant differences between clinically relevant groups. These tests are generally more powerful with larger sample sets. While statistics can be extremely useful in screening for candidate biomarkers, they can also be misleading, so it is important to further evaluate statistically significant peaks and select the most robust peaks.

P Values

ProteinChip data manager software includes several univariate statistical techniques for analysis of individual peak clusters. The most commonly used P value calculations are the Mann-Whitney (2 groups) or Kruskal-Wallis (>2 groups) nonparametric tests. The software also includes a Wilcoxon test, which can be used for paired analyses (with samples collected before and after treatment from the same patient). Average replicates prior to conducting any statistical analysis.

Low P values may indicate a significant difference in mean expression levels for a particular protein (peak cluster). Though P values of <0.05 are generally considered significant, $P < 0.01$ is a reasonable starting point for more robust markers. However, no formal P value threshold holds true for all experiments. The numbers of samples and candidate biomarkers and other statistical principles, such as false discovery rate, all play a major role in determining the appropriate P value for a given experiment. Input from a biostatistician will greatly assist in formalizing the correct P value threshold. In addition, other statistical tools, such as the receiver-operator characteristic curves described below, should be used in conjunction with P values to obtain proper statistical evaluation of biomarker candidates.

Receiver-Operator Characteristic (ROC) Curves

ROC curves are useful tools that can be used to estimate clinical sensitivity and specificity. Use the ROC plots to compare two cohorts and to plot sensitivity vs. [1 – specificity]. Whereas a calculated area under the ROC plot (AUC) of 0.5 indicates no separation between cohorts, an AUC of 1.0 indicates 100% sensitivity and specificity. AUC is normally reported as a value of >0.5, which indicates the peak is upregulated in the selected positive group. During screening, also examine values of <0.5, which represent peaks that are downregulated in the selected positive group.

To perform a standard workflow for univariate analysis:

1. Calculate P values and AUC using ProteinChip data manager software.
2. Evaluate peak clusters with low P values and good AUC values, which indicate separation of relevant cohorts. As with P values, appropriate AUC cutoffs depend on the clinical question and patient population; however, an AUC of >0.8 (or <0.2 for downregulated peaks) is a good starting point.
3. Curate selected peak clusters. Look for well-resolved peaks with good signal-to-noise ratios in a significant number of spectra, and relabel any peaks that were mislabeled during automatic peak detection using the “edit cluster” feature.
4. Recalculate P values and AUC values after peak relabeling, ensuring that spectra from replicate samples are averaged together.

Evaluate peak clusters with good potential as single biomarkers based on univariate statistical analysis. Specifically, evaluate the peak quality, percent difference between groups, and consistency across conditions or patients. You may also select these peaks for purification, identification, or validation. Alternatively, analyze a selection of peak clusters using multivariate statistical analysis tools to develop classification algorithms with improved clinical sensitivity and specificity over single markers alone.

Perform Multivariate Statistical Analyses

In many cases, it is difficult to find a single marker that accurately distinguishes one class of samples from another. This is particularly true in human trials, in which patient-to-patient variability is high. In these cases, use multivariate analysis to analyze patterns of protein expression and develop algorithms to improve classification. Collaborate with biostatisticians throughout study design and during the complex phases of statistical data analysis, especially when moving to multivariate analyses.

SELDI, like most genomics and proteomics methods, generates high-dimensional data, in which the number of data points (peaks) normally far exceeds the number of samples. Therefore, data must be analyzed carefully to avoid false discovery and over fitting of multivariate models. Construction of robust multivariate models generally requires large sample numbers. Consult with a biostatistician to determine the minimum number of samples required for multivariate analysis for a specific clinical question and patient population.

Multivariate analyses can be grouped into supervised or unsupervised techniques.

Unsupervised Techniques

Use these techniques to determine natural groupings (clusters) within data without using prior sample or patient information. Visualization of clustering analyses can reveal desired (disease diagnosis, drug treatment efficacy, adverse drug effect) or undesired (preanalytical or analytical bias) groupings within the data. Two clustering techniques, hierarchical clustering and principal components analysis (PCA), can reduce the dimensionality of ProteinChip SELDI peak cluster data for easier visualization; both are available in ProteinChip data manager software.

- Hierarchical clustering analyses generate heat map dendrograms in a fashion analogous to microarray data analysis of gene expression studies (Eisen et al. 1998)

- PCA results generate two- and three-dimensional graphs, displaying a list of the principal components along with their eigen values

Both techniques may be used with all detected peak clusters to visualize differences between cohorts. Simpler models can be generated by using a smaller set of peak clusters selected based on significant P values and AUC values.

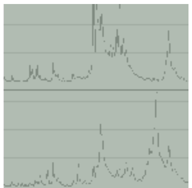
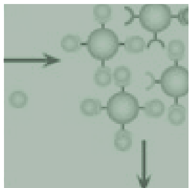
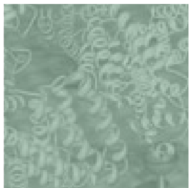
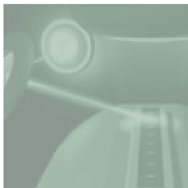
Supervised Techniques

Use these techniques, also known as classification tools, for determining and testing diagnostic algorithms. Analyze peak cluster data exported from ProteinChip data manager software with ProteinChip pattern analysis software, which uses classification tree analyses to correlate multiple biomarkers with specific phenotypes and improve classification accuracy. The output is an easy-to-interpret decision tree that uses a small panel of markers with defined splitting rules for classification. Typically, several decision tree models from the discovery phase are selected for further testing and refinement in the validation phase. In addition, exported peak cluster data can be analyzed by a variety of commercially available classifier algorithms, including partial least squares, maximum likelihood, K nearest neighbor, and support vector machines.

General guidelines for multivariate analysis, independent of the classification tool, include the following:

- Determine and implement the best method for treating missing data, such as spectra eliminated due to extreme normalization factors, for your specific classification tool
- Using peak clusters created by ProteinChip data manager software, perform feature selection techniques provided by multivariate classification tools to define important classifier peaks that may be missed by the simple P value and AUC analyses
- Use resampling techniques such as statistical cross-validation or bootstrapping to estimate the test error rate of the classification algorithm
- Evaluate and compare the performance of different classification algorithms by plotting ROC curves and calculating AUC values for each algorithm in a fashion analogous to ROC plots of single peak clusters
- Validate and refine multivariate classification models on an independent data set

Appendix A: Methods

Appendix A: Methods

The Bio-Rad Biomarker Research Centers have developed a series of standard experimental operating procedures that are based on a number of successful ProteinChip SELDI projects using a variety of sample types. This section outlines standard procedures for array processing, for fractionation and analysis of serum or plasma, and for analysis of cerebrospinal fluid (CSF) and urine.

General Protocol for Processing ProteinChip Arrays

This protocol describes general guidelines to be followed for array preparation using a ProteinChip bioprocessor (catalog #C50-30011). Always do the following:

- Randomize sample placement across arrays. Ensure that samples representing different clinical groups are equally represented on each array and bioprocessor
- Analyze multiple aliquots of an appropriate reference sample within each bioprocessor to monitor sample preparation for quality control purposes. Examples of reference samples include human serum (catalog #C10-00006), CSF, urine, or other sample selected to match the type of sample being profiled
- Prepare a pooled sample using a subset of the actual study samples for optimization of the data acquisition settings
- Consider automating array preparation steps by using a liquid-handling workstation (such as those available from Beckman, Tecan, Hamilton, etc.) to further reduce bias and improve reproducibility
- Take precautions to minimize bubble formation as described in Part III and below

Procedure

Array Pretreatment

Pretreat the ProteinChip array surfaces using different buffers depending on the array type.

Recommended binding and wash buffers for ProteinChip arrays.

ProteinChip Array	Low-Stringency Binding and Wash Buffer	High-Stringency Binding and Wash Buffer
IMAC30	100 mM $NaPO_4$, pH 7.0, 0.5 M NaCl	
H50	10% Acetonitrile, 0.1% trifluoroacetic acid (TFA), pH 1.9	
CM10	100 mM $NaCH_3CO_2$, pH 4.0	50 mM HEPES, pH 7.0
Q10	50 mM Tris-HCl, pH 9.0	100 mM MES, pH 6.5

ProteinChip IMAC30 arrays — Charge ProteinChip IMAC30 arrays with Cu^{+2} or Ni^{+2}.

1. Place the arrays in a ProteinChip bioprocessor and load 50 µl 100 mM $CuSO_4$ or 50 µl 100 mM $NiSO_4$ onto each spot on the ProteinChip IMAC30 array. Monitor bioprocessors for bubbles and take precautions to minimize bubble formation. If possible, centrifuge the bioprocessor for 1 min at 250 x g to ensure the buffer is directly in

contact with the array surface. Shake 5 min at room temperature. Remove the $CuSO_4$ or $NiSO_4$ after shaking. Rinse with deionized water.

2. When using $CuSO_4$, load 50 µl 100 mM $NaCH_3CO_2$, pH 4.0 onto each spot on the ProteinChip IMAC30 array. Shake 5 min at room temperature. Remove the $NaCH_3CO_2$ after shaking. Rinse with deionized water.

ProteinChip H50 arrays — Place the arrays in a ProteinChip bioprocessor. Add 50 µl 50% acetonitrile in deionized water to each bioprocessor well. Shake the arrays for 5 min. Remove the wash solution after shaking.

ProteinChip CM10 and Q10 arrays — Place the arrays in a ProteinChip bioprocessor. No pretreatment is required.

Array Equilibration
Add 150 µl appropriate array binding buffer (see table) to each well. Monitor bioprocessors for bubbles and take precautions to minimize bubble formation as above. Shake 5 min at room temperature. Remove buffer. Add 150 µl binding buffer into each well. Shake 5 min at room temperature. Remove buffer before adding samples.

Sample Dilution and Binding
Dilute replicate aliquots of each sample by first adding the buffer to each well and then adding the sample to each well. Monitor bioprocessors for bubbles and take precautions to minimize bubble formation as above. Shake 60 min (this time may require optimization within the range of 30–120 min) at room temperature. Remove sample and buffer.

Note: The details of this dilution and binding step may require optimization for different sample types, some of which are described in the following protocols.

Array Washing
Add 150 µl appropriate array wash buffer (same as binding buffer, see Table) into each well. Shake 5 min at room temperature. Remove the buffer after shaking. Repeat this wash step two more times for a total of three washes. Rinse twice with deionized water (except H50 arrays, which do not require a water rinse). Remove the top of the bioprocessor and allow spots to air-dry for 20–30 min (use a consistent drying time for all arrays).

Matrix Preparation and Application
Use sinapinic acid (SPA) for general analysis of peptides and proteins of all molecular weights. Alternatively, use α-cyano-4-hydroxycinnamic acid (CHCA) for enhanced detection of low-mass peptides. If using both matrices, prepare duplicate sets of arrays: add SPA to one set, and CHCA to the other set. Do not add both SPA and CHCA on the same array.

To prepare ProteinChip SPA matrix:

1. Add 400 µl 50% acetonitrile, 0.5% TFA to the vial containing the SPA matrix. Shake 5 min at room temperature.
2. Apply 1.0 µl prepared matrix to each spot. Allow spots to air-dry for 10–20 min (use a consistent drying time for all arrays). Apply 1.0 µl to each spot again. Allow spots to air-dry for at least 60 min.

To prepare ProteinChip CHCA matrix:

1. Add 200 µl 50% acetonitrile, 0.25% TFA to the tube containing the CHCA matrix. Shake 5 min at room temperature. Centrifuge at 10,000 x g for 1 min at room temperature. Remove supernatant and dilute with four volumes of 50% acetonitrile, 0.25% TFA.
2. Apply 1.0 µl prepared matrix to each spot. Allow spots to air-dry for 10–20 min (use a consistent drying time for all arrays). Apply 1.0 µl to each spot again. Allow spots to air-dry for at least 60 min.

Analyze the arrays within a few days of preparation. Store prepared arrays in a dark dessicator at room temperature.

ProteinChip SELDI Analysis of Serum or Plasma

This protocol describes a general workflow for serum or plasma analysis. Plasma and serum samples are fractionated, and the resulting fractions are then analyzed on different array types using a ProteinChip bioprocessor. For sample fractionation, use ProteoMiner™ technology when sample volumes of 1 ml are available, or use strong anion exchange chromatography when sample volumes are limited (20 µl).

Procedure

Sample Preparation

Perform fractionation using either ProteoMiner technology or strong anion exchange chromatography.

- For large sample volumes (1 ml), use the ProteoMiner sequential elution kit (catalog #163-3002). Follow the detailed instructions provided with the kit
- For small sample volumes (20 µl), use strong anion exchange chromatography and the ProteinChip serum fractionation kit (catalog #K10-00007), which uses a 96-well microplate format for high-throughput processing, or the lower throughput ProteinChip Q spin columns (catalog #C54-00017). Follow the detailed instructions provided with the corresponding kit

Array Processing

Follow the steps described in the General Protocol for Processing ProteinChip Arrays section, profiling each of the fractions in duplicate or triplicate on the ProteinChip arrays listed in the table. The manner of sample dilution, types of arrays, and binding buffers used depend on the method of fractionation used.

- When using ProteoMiner technology for fractionation, profile each of the four resulting fractions on four different array types. To monitor the expression levels of high-abundance proteins, also analyze the original sample or the flowthrough fraction on all four array types
- When using strong anion exchange chromatography for fractionation, profile all six resulting fractions on three different array types. If needed to reduce the number of fractions, omit fraction 2 and combine fractions 5 and 6. Also, omit fraction 2 if samples are contaminated with hemoglobin, which elutes primarily in this fraction

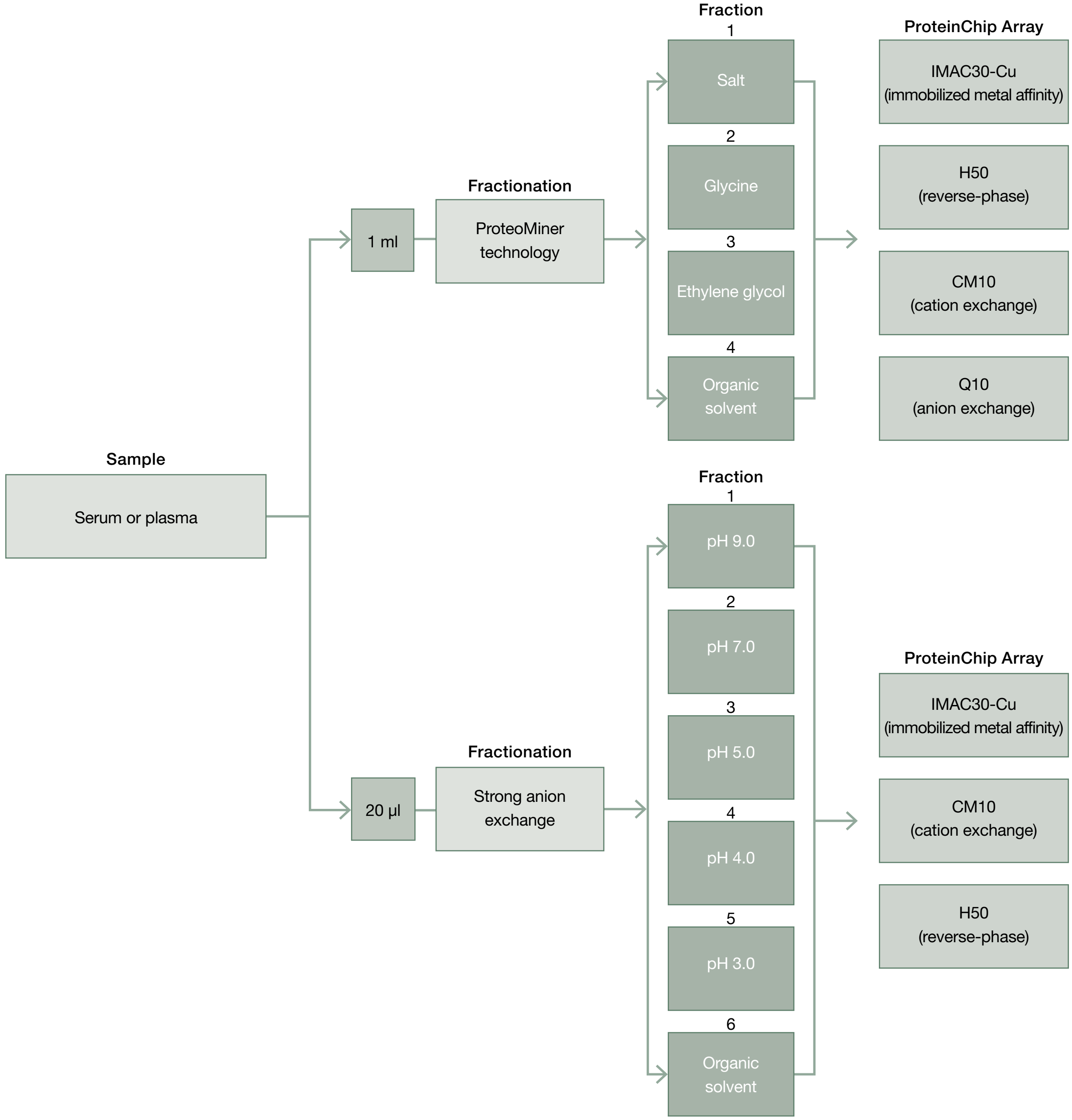
As a first choice of matrix, use SPA. If additional proteome coverage is required, prepare an additional set of arrays using CHCA as the matrix.

Data Collection and Data Analysis

No single set of data collection or data analysis protocols and parameters is suitable for all samples and all experimental needs. Follow the recommendations detailed in the Data Collection and Data Analysis sections in Part III of this guide for establishing the most appropriate procedures.

Dilution and ProteinChip arrays to be used with fractionated plasma and serum samples. For all array types, use the appropriate low-stringency binding buffer (see table in the General Protocol for Processing ProteinChip Arrays section).

Fractionation Method	1:10 Dilution	Array Type
ProteoMiner technology	45 µl binding buffer + 5 µl each fraction	IMAC30-Cu H50 CM10 Q10
Strong anion exchange	90 µl binding buffer + 10 µl each fraction	IMAC30-Cu H50 CM10



Analysis workflow. Each serum or plasma sample is fractionated, and the resulting fractions are then analyzed on different ProteinChip array chemistries. Note that for ProteoMiner technology, the original sample or flowthrough can also be analyzed on all four arrays in order to monitor expression levels of high-abundance proteins.

ProteinChip SELDI Analysis of Cerebrospinal Fluid

Cerebrospinal fluid (CSF) is roughly 20 times more dilute than serum or plasma; therefore, fractionation is generally not necessary. In this protocol, CSF samples are diluted 1:10 for direct binding onto two sets of five different ProteinChip arrays: one set is analyzed using CHCA as the matrix and the other with SPA.

Procedure

Sample Preparation

Collect CSF samples in polypropylene tubes and gently mix the samples. Centrifuge the samples at 2,000 x g for 10 min at 4°C to remove cells and other insoluble material. Aliquot the samples and freeze and store them at –80°C. Samples should not contain more than 500 erythrocytes/µl based on blood cell counts. Thaw frozen CSF samples and centrifuge them at 20,000 x g for 5 min at 4°C to pellet debris. Create a pool of CSF from all or a subset of experimental samples, and use the pooled sample for optimization of instrument settings and as the quality control sample.

Array Processing

Follow the steps described in the General Protocol for Processing ProteinChip Arrays section, profiling each sample in duplicate or triplicate on the five different types of ProteinChip arrays listed below. Dilute samples 1:10 by adding 45 µl of appropriate binding buffer to each well and then adding 5 µl of sample per well.

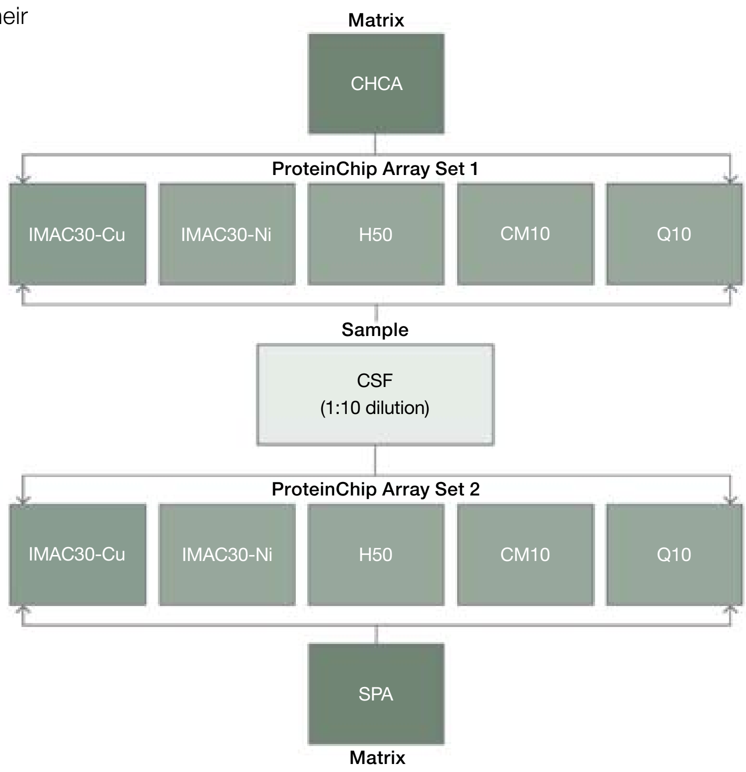
Use the following five array chemistries with their corresponding low-stringency buffers:

- IMAC30-Cu array
- IMAC30-Ni array
- H50 array
- CM10 array
- Q10 array

Prepare two sets of arrays: one using SPA as the matrix, and a second using CHCA. If sample volume is limited, prepare one set of arrays and use SPA as the matrix.

Data Collection and Data Analysis

No single set of data collection or data analysis protocols and parameters is suitable for all samples and all experimental needs. Follow the recommendations detailed in the Data Collection and Data Analysis sections of Part III of this guide for establishing the most appropriate procedures.



Analysis workflow. Each CSF sample is diluted 1:10 for direct binding on five different ProteinChip array chemistries. Two sets of arrays are prepared with different matrix preparations: one set is analyzed with CHCA, and the other with SPA.

ProteinChip SELDI Analysis of Urine

Urine samples are more dilute than most other sample types. This protocol includes a serial sample incubation to maximize the amount of protein binding to the array surface.

Procedure

Sample Preparation

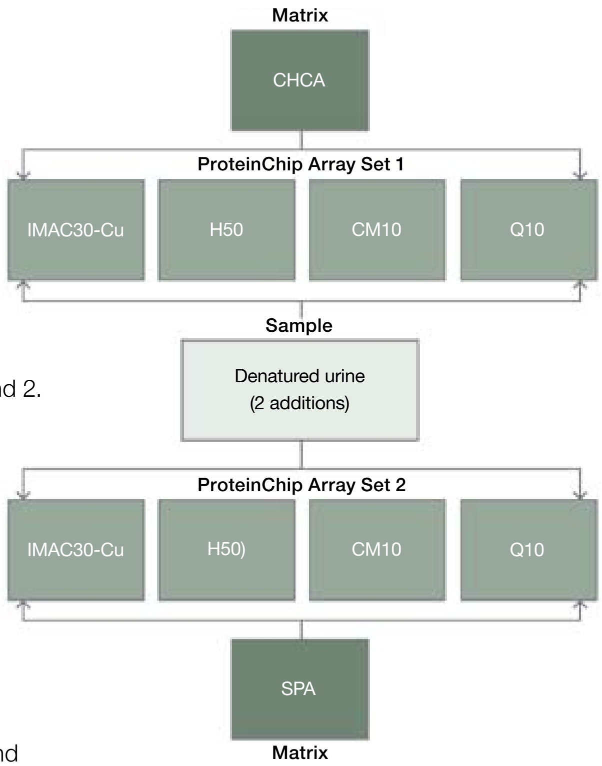
Collect samples following a standardized protocol. If possible, collect samples over a 24 hr period; otherwise, use the first or second morning void samples. Aliquot the samples and freeze and store them at –80°C. Thaw the frozen urine samples and centrifuge them at 20,000 x g for 5 min at 4°C to pellet debris. Because the protein concentration of urine samples can be variable, pool urine aliquots from all or a subset of samples for optimization of instrument settings and for use as the quality control sample. Denature the samples by mixing 110 µl each supernatant with 175 µl 9 M urea, 2% CHAPS, 50 Mm Tris-HCl, pH 9.0 and incubating for 30 min at 4°C.

Array Processing

Follow the steps described in the General Protocol for Processing ProteinChip Arrays section, profiling each sample in duplicate or triplicate on the following ProteinChip arrays:

- IMAC30-Cu array, low-stringency buffer
- H50 array, low-stringency buffer
- CM10 array, low-stringency buffer
- Q10 array, high-stringency buffer

1. Dilute 25 µl denatured urine sample in 175 µl of the appropriate binding buffer and apply to pre-equilibrated ProteinChip arrays.
2. Allow each sample to bind to the array surface for 30 min at room temperature.
3. Remove the sample from the array and repeat steps 1 and 2.
4. Remove the sample from the array and wash each array three times with the appropriate binding buffer. Rinse twice with water (except H50 arrays, which do not require a water rinse).
5. Prepare two sets of arrays: one using SPA as the matrix, and a second using CHCA. If sample volume is limited, prepare one set of arrays and use SPA as the matrix.

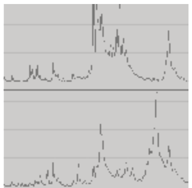
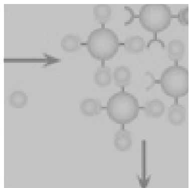
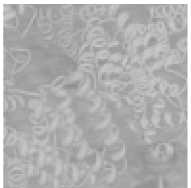


Analysis workflow. Each urine sample is profiled on four different ProteinChip array chemistries. Two sets of arrays are prepared with different matrix preparations: one set is analyzed with CHCA, and the other with SPA.

Data Collection and Data Analysis

No single set of data collection or data analysis protocols and parameters is suitable for all samples and all experimental needs. Follow the recommendations detailed in the Data Collection and Data Analysis sections of Part III of this guide for establishing the most appropriate procedures.

Appendix B: References and Related Reading


Appendix B: References and Related Reading

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Bio-Rad Bulletins

Bulletin	Title
3145	Strategies for Protein Sample Preparation
10008220	ProteinChip® SELDI System: Reader Guide, Personal Edition and Enterprise Edition
10008221	ProteinChip SELDI System: Applications Guide, Volume 1
10008222	ProteinChip SELDI System: Applications Guide, Volume 2
10008226	ProteinChip Data Manager Software Operation Manual

Related Reading

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